MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

COMPARATOR, MAGNITUDE, DIGITAL, KEY CODE HFB

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic magnitude comparator module.

1.2 Classification. This specification covers two classes of modules.

Class I and Class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.162 failures per million operating hours for class I or 0.502 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class two module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HFB.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/300-1 for class I and M28787/300-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

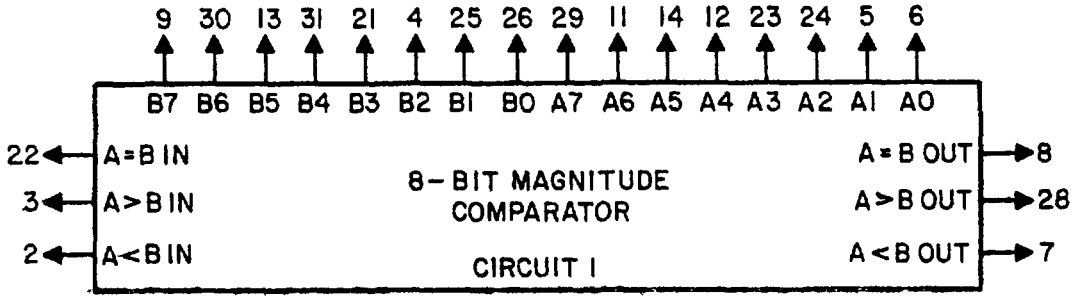
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

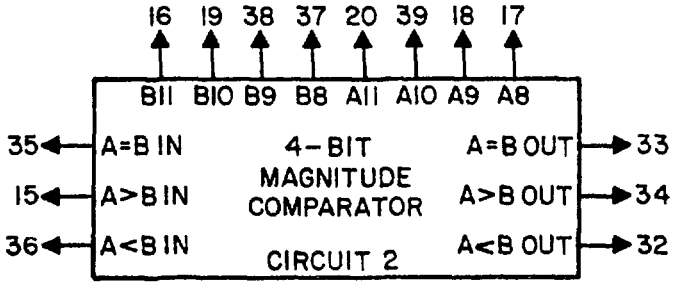
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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1 ← V_{CC} (5.0±0.5 V dc)
10 ← GND (0V REF)
27 ← (NOT USED)
40 ← (NOT USED)

A0-A11, B0-B11 - COMPARING INPUTS
A=B IN, A>B IN, A<B IN - CASCADING INPUTS
A=B OUT, A>B OUT, A<B OUT - OUTPUTS

FIGURE 1. HFB module function diagram, magnitude comparator.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1,A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B6	V_{OUT} (low loaded)	IV	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
C1-C26	Input breakdown current	V	S	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	0.0	0.3	mA
C27-C30	Input breakdown current	V	S	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	mA
D1-D6	Output leakage current	VI	X	0	1	0	1	0	1	0	1	0	1	mA
E1-E6	Output short circuit current	VI	X	-30	-130	-30	-130	-30	-130	-30	-130	-30	-130	mA
F1-F30	Input clamp diode voltage	VII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G26	Input current, high level	VIII	X	0	60	0	60	0	60	0	60	0	60	µA
G27-G30	Input current, high level	VIII	X	0	20	0	20	0	20	0	20	0	20	µA
H1-H26	Input current, low level	VIII	X	-0.10	-1.32	-0.10	-1.32	-0.10	-1.32	-0.10	-1.32	-0.10	-1.32	mA
H27-H30	Input current, low level	VIII	X	-0.04	-0.44	-0.04	-0.44	-0.04	-0.44	-0.04	-0.44	-0.04	-0.44	mA
J1-J54	Functional compare	IX	X	← See Table IX →										
K1-K6	Output voltage, high level	IX	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
L1-L6	Output voltage, low level	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
M	Power supply current	X	S	15	60	15	60	15	60	15	60	15	60	mA
N1-N6	Transition time (t_{TLH}), C_L = 50 pF	XI	S	5	50	5	50	5	55	5	50	5	58	ns
N7-N12	Transition time (t_{TLH}), C_L = 150 pF	XI	Q	10	80	10	80	10	90	10	80	10	96	ns
P1-P6	Transition time (t_{THL}), C_L = 50 pF	XI	S	3	18	3	18	3	20	3	18	3	21	ns
P7-P12	Transition time (t_{THL}), C_L = 150 pF	XI	Q	6	40	6	40	6	42	6	40	6	43	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q4	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	5	52	5	52	5	56	5	52	6	59	ns
Q5	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	4	48	4	48	4	52	4	48	4	55	ns
Q6-Q9	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	26	2	26	2	28	2	26	3	29	ns
Q10	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	2	24	2	24	2	26	2	24	2	27	ns
Q11-Q14, Q19-Q22, Q27-Q30, Q35-Q38	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	4	40	4	40	4	42	4	40	4	43	ns
Q15-Q18, Q23-Q26, Q31-Q34, Q39-Q42	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	4	49	4	49	5	51	4	49	5	52	ns
Q43-Q50, Q59-Q66, Q75-Q82, Q91-Q98	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	8	80	8	80	8	84	8	80	8	89	ns
Q51-Q58, Q67-Q74, Q83-Q90, Q99-Q106	Propagation delay time (t_{PLH}), C_L = 50 pF	XII	S	9	98	9	98	10	102	9	98	10	105	ns
Q107-Q110	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	7	72	7	72	7	76	7	72	8	79	ns
Q111	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	6	68	6	68	7	72	6	68	7	75	ns
Q112-Q115	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	3	36	3	36	3	38	3	36	4	39	ns
Q116	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	3	34	3	34	3	36	3	34	3	37	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q117-Q120, Q125-Q128, Q133-Q136, Q141-Q144	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	5	50	5	50	5	52	5	50	5	53	ns
Q121-Q124, Q129-Q132, Q137-Q140, Q145-Q148	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	5	59	5	59	6	61	5	59	6	62	ns
Q149-Q156, Q165-Q172, Q181-Q188, Q197-Q204	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	10	100	10	100	10	104	10	100	10	107	ns
Q157-Q164, Q173-Q180, Q189-Q196, Q205-Q212	Propagation delay time (t_{PLH}), C_L = 150 pF	XII	Q	11	118	11	118	12	122	11	118	12	125	ns
R1-R4	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	4	42	4	42	4	46	4	42	5	49	ns
R5	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	6	60	6	60	6	64	6	60	6	67	ns
R6-R9	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	2	21	2	21	2	23	2	21	2	24	ns
R10	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	3	30	3	30	3	32	3	30	3	33	ns
R11-R14, R19-R22, R27-R30, R35-R38	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	3	34	3	34	3	36	3	34	3	37	ns
R15-R18, R23-R26, R31-R34, R39-R42	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	4	49	4	49	5	51	4	49	5	52	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R43-R50, R59-R66, R75-R82, R91-R98	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	6	68	6	68	7	72	6	68	7	75	ns
R51-R58, R67-R74, R83-R90, R99-R106	Propagation delay time (t_{PHL}), C_L = 50 pF	XII	S	9	98	9	98	10	102	9	98	10	105	ns
R107-R110	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	6	62	6	62	6	66	6	62	7	69	ns
R111	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	8	80	8	80	8	84	8	80	8	87	ns
R112-R115	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	3	31	3	31	3	33	3	31	3	34	ns
R116	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	4	40	4	40	4	42	4	40	4	43	ns
R117-R120, R125-R128, R133-R136, R141-R144	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	4	44	4	44	4	46	4	44	4	47	ns
R121-R124, R129-R132, R137-R140, R145-R148	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	5	59	5	59	6	61	5	59	6	62	ns
R149-R156, R165-R172, R181-R188, R197-R204	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	8	88	8	88	9	92	8	88	9	95	ns
R157-R164, R173-R180, R189-R196, R205-R212	Propagation delay time (t_{PHL}), C_L = 150 pF	XII	Q	11	118	11	118	12	122	11	118	12	125	ns

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See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5° C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S	Power supply filter capacitance	XIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
T1-T6	Output crosstalk, high level	XIV	Q	0	400	0	400	0	400	0	400	0	400	mV
U1-U6	Output crosstalk, low level	XIV	Q	0	500	0	500	0	500	0	500	0	500	mV
V	Power supply transient amplitude	XV	Q	0	500	0	500	0	500	0	500	0	500	mV
W1-W30	Input capacitance	XVI	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition S, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation (outputs not loaded)	0.33	0.33	W
Input current (A, B & A=B)			
Low level (V_{IL}=0.0 V)	-0.10 min, -1.32 max	-0.10 min, -1.32 max	mA
Low level (V_{IL}=0.4 V)	-0.10 min, -1.2 max	-0.1 min, -1.2 max	mA

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Input current			
High level (V_{IH}=2.4 V)	0 min, 60 max	0 min, 60 max	µA
High level (V_{IH}=5.5V)	0.0 min, 0.3 max	0.0 min, 0.3 max	mA
Output drive current			
Low level (V_{OL}=0.4 V)	4	4	mA
High level (V_{OH}=2.5 V)	-400	-400	µA
Input current (A>B & A<B			
Low level (V_{IL}=0.0 V)	-0.04 min, -0.44 max	-0.04 min, -0.44 max	mA
(V_{IL}=0.4 V)	-0.04 min, -0.40 max	-0.04 min, -0.40 max	mA
High level (V_{IH}=2.4 V)	0 min, 20 max	0 min, 20 max	µA
(V_{IH}=5.5 V)	0 min, 0.1 max	0 min, 0.1 max	mA

6.5 Input to output relationship. Each magnitude comparator module has the input to output relationship of table II for positive logic.

6.5.1 Parallel outputs. Paralleling of outputs is not permissible.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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(Project 5963-N464)

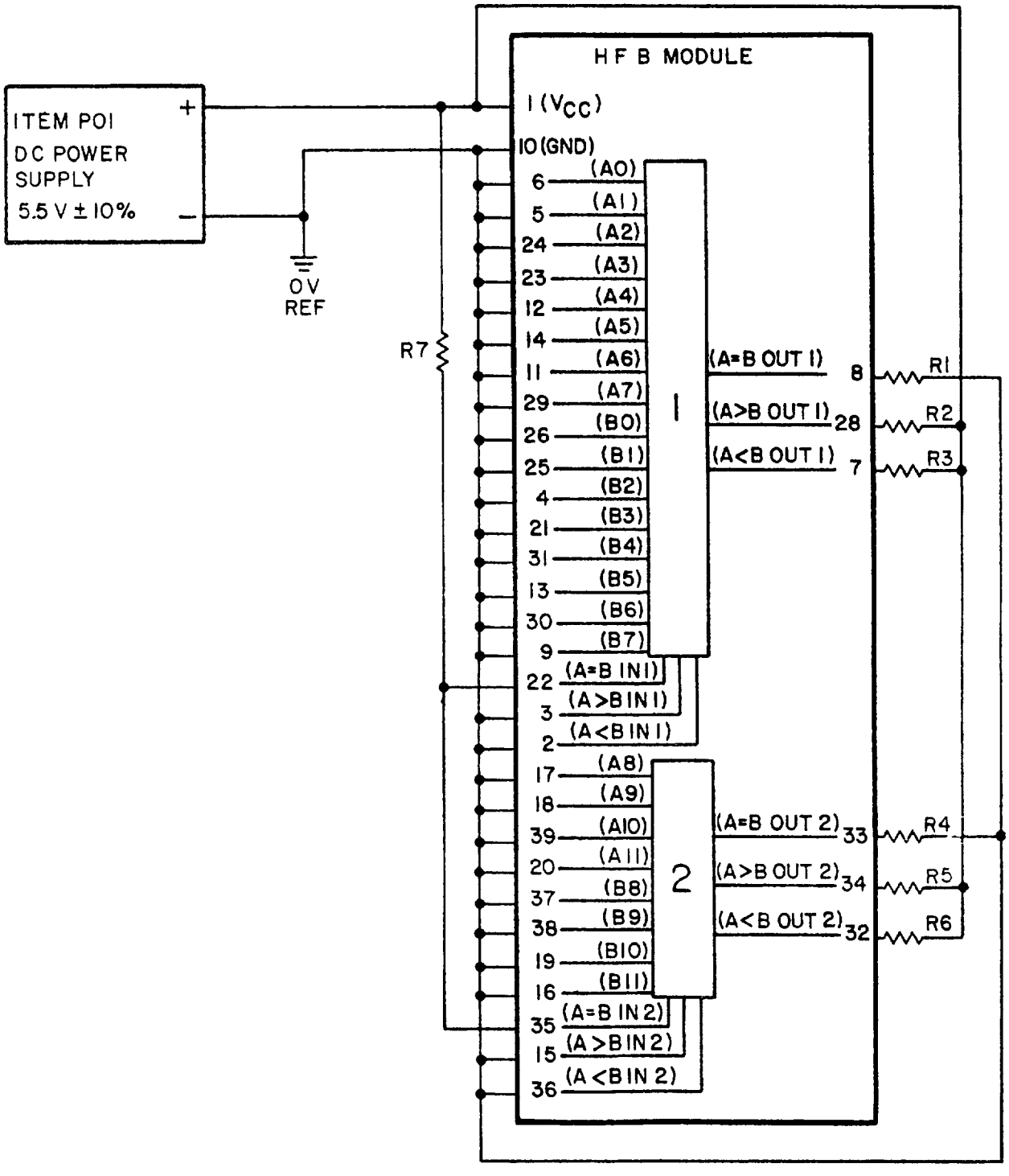
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Table II. Input to output relationship. 1/

Comparing inputs												Cascading inputs			Outputs		
A11,B11	A10,B10	A9,B9	A8,B8	A7,B7	A6,B6	A5,B5	A4,B4	A3,B3	A2,B2	A1,B1	A0,B0	A>B	A<B	A=B	A>B	A<B	A=B
A11>B11	X	X	X									X	X	X	H	L	L
A11<B11	X	X	X									X	X	X	L	H	L
A11=B11	A10>B10	X	X									X	X	X	H	L	L
A11=B11	A10<B10	X	X									X	X	X	L	H	L
A11=B11	A10=B10	A9>B9	X									X	X	X	H	L	L
A11=B11	A10=B10	A9<B9	X									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8>B8									X	X	X	H	L	L
A11=B11	A10=B10	A9=B9	A8<B8									X	X	X	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									H	L	L	H	L	L
A11=B11	A10=B10	A9=B9	A8=B8									L	H	L	L	H	L
A11=B11	A10=B10	A9=B9	A8=B8									L	L	H	L	L	H
				A7>B7	X	X	X	X	X	X	X	X	X	X	H	L	L
				A7<B7	X	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6>B6	X	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6<B6	X	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5>B5	X	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5<B5	X	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4>B4	X	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4<B4	X	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3>B3	X	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3<B3	X	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2>B2	X	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2<B2	X	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1>B1	X	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1<B1	X	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0>B0	X	X	X	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0<B0	X	X	X	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	H	L	L	H	L	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	H	L	L	H	L
				A7=B7	A6=B6	A5=B5	A4=B4	A3=B3	A2=B2	A1=B1	A0=B0	L	L	H	L	L	H

1/ H = high level voltage, L = Low level voltage, X = don't care.

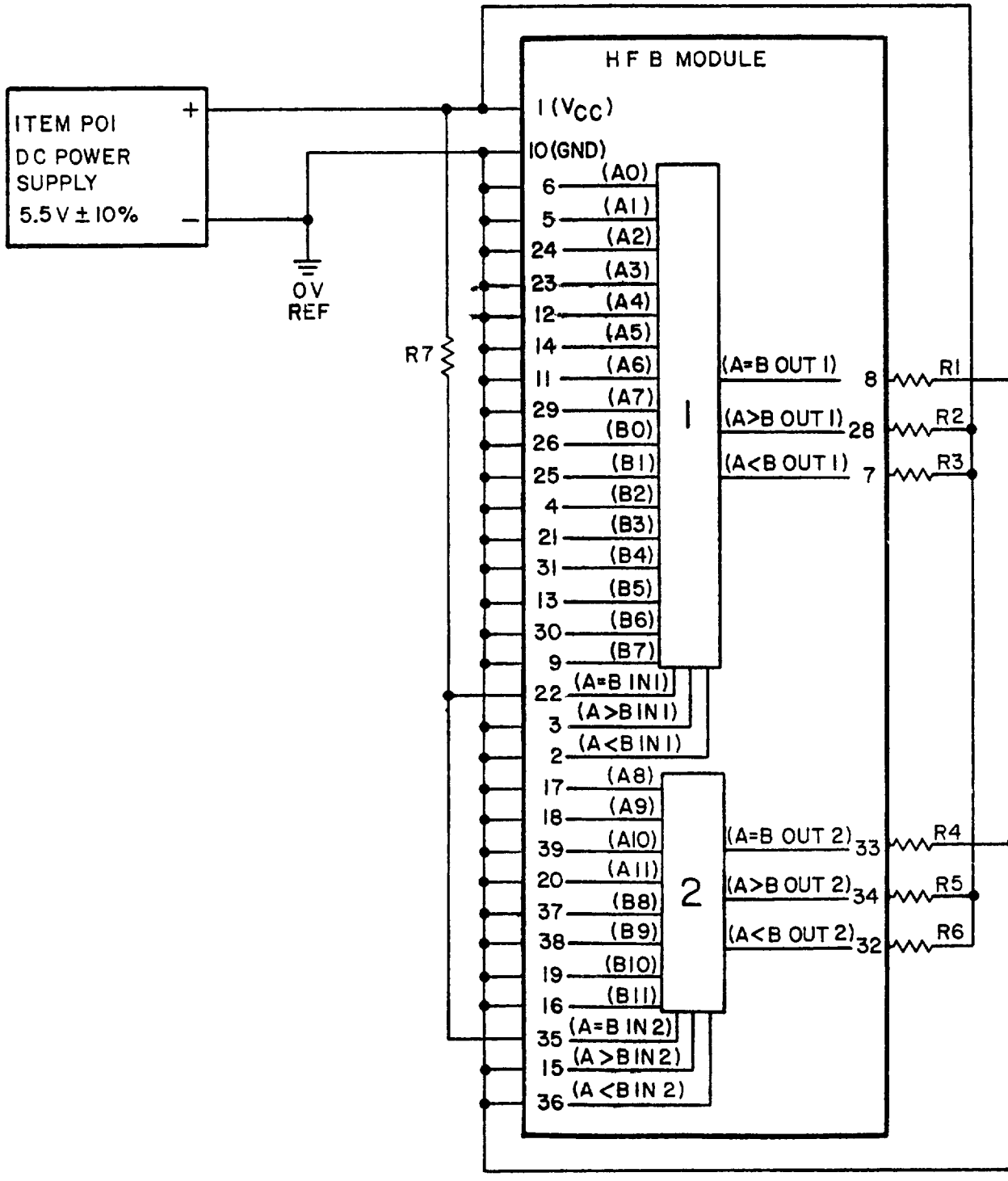
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NOTES:
1. R1 = R4 = 6.19 kΩ ±1%, 1/8 W.
2. R2 = R3 = R5 = R6 = 1.27 kΩ ±1%, 1/4 W.
3. R7 = 1 kΩ ±10%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R4 = 6.19 kΩ ±1%, 1/8 W.
2. R2 = R3 = R5 = R6 = 1.27 kΩ ±1%, 1/4 W.
3. R7 = 1 kΩ ±10%, 1/8 W.

FIGURE 3. Component temperature test circuit.

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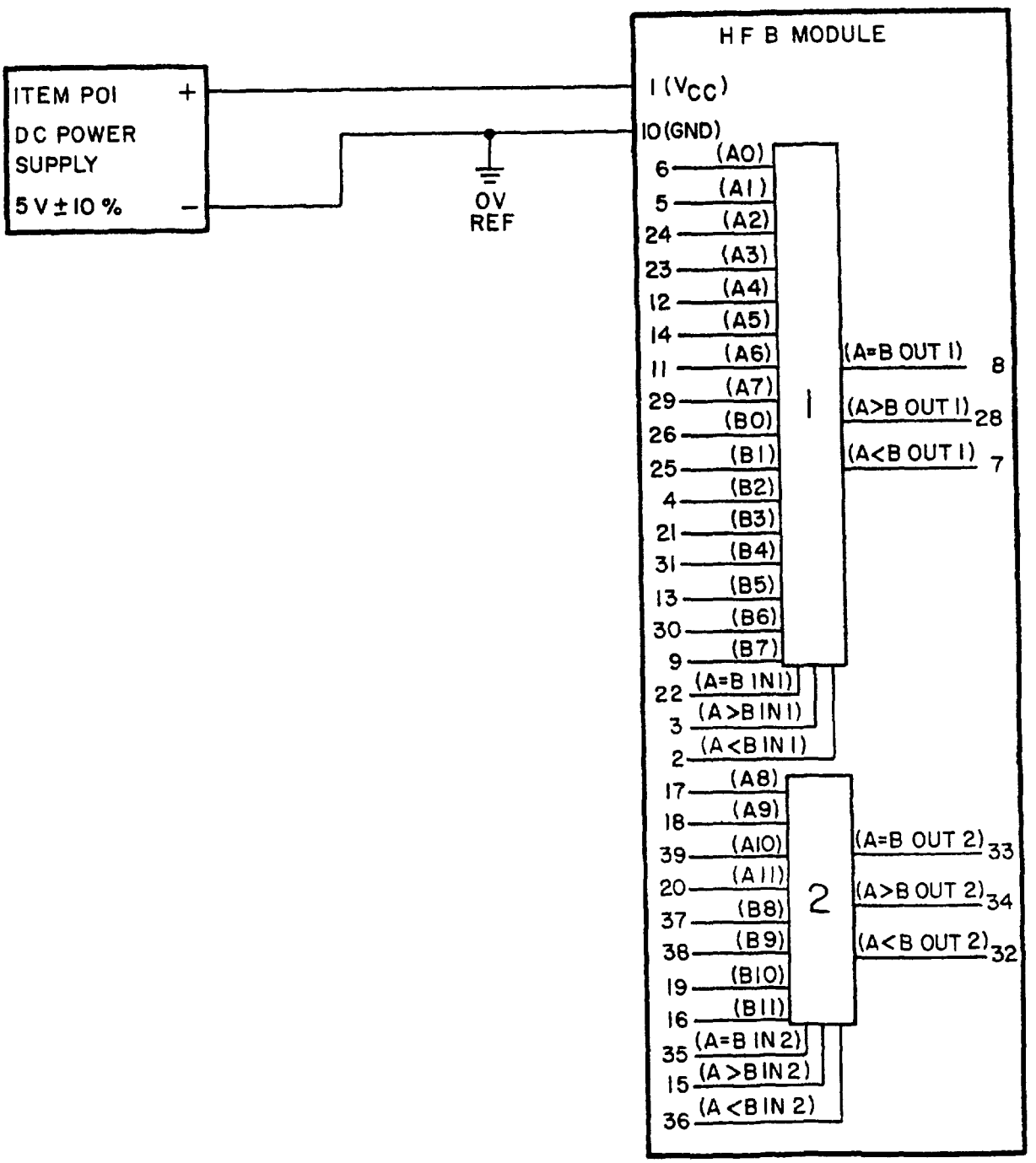


FIGURE 4. Vibration test circuit.

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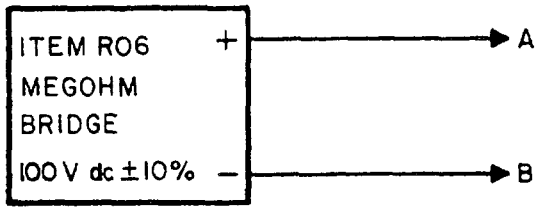
TABLE III. Isolation test.

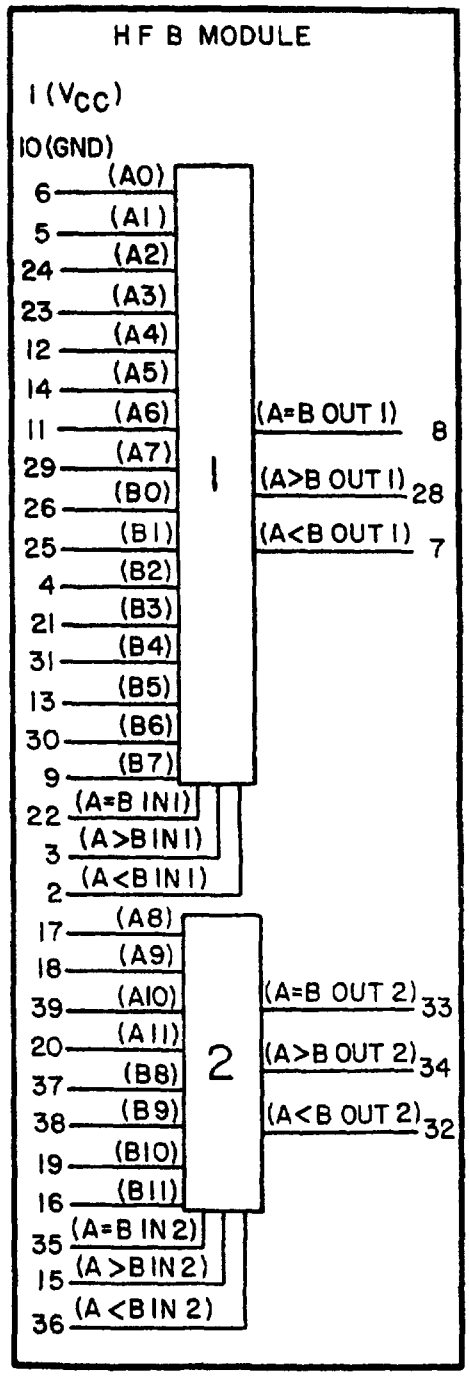
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation, frame to module pins	All pins	Frame	10	∞	MΩ	5
A2	Isolation, used to unused pins	All pins except 27,40	27,40	10	∞	MΩ	5

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	A>B OUT 1 V_{OUT} (low loaded)	28	29	0.01	0.80	V	6
B2	A<B OUT 1 "	7	9	"	"	"	"
B3	A=B OUT 1 "	8	9	"	"	"	"
B4	A>B OUT 2 "	34	20	"	"	"	"
B5	A<B OUT 2 "	32	16	"	"	"	"
B6	A=B OUT 2 "	33	16	"	"	"	"

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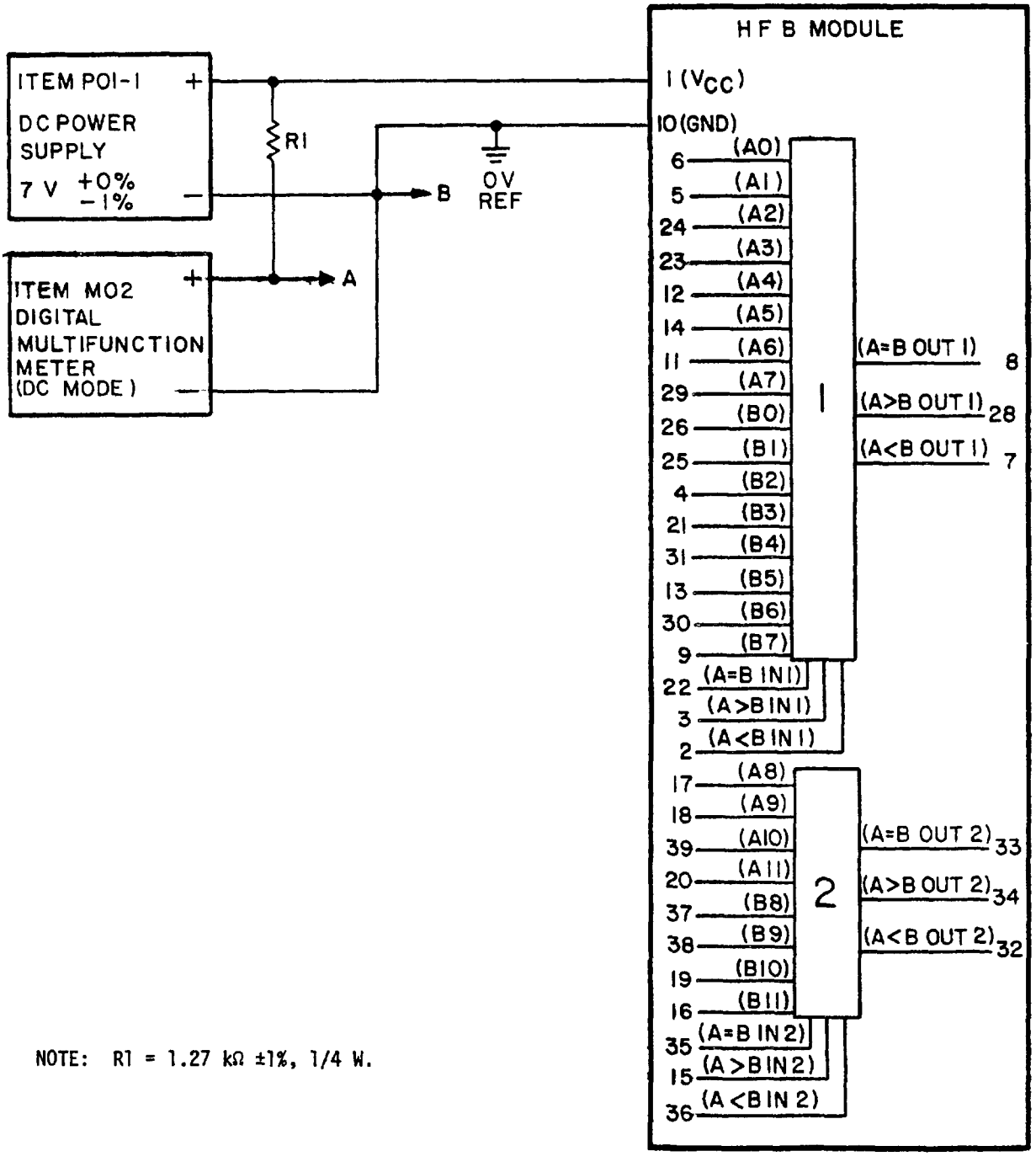
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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NOTE: R1 = 1.27 kΩ ±1%, 1/4 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

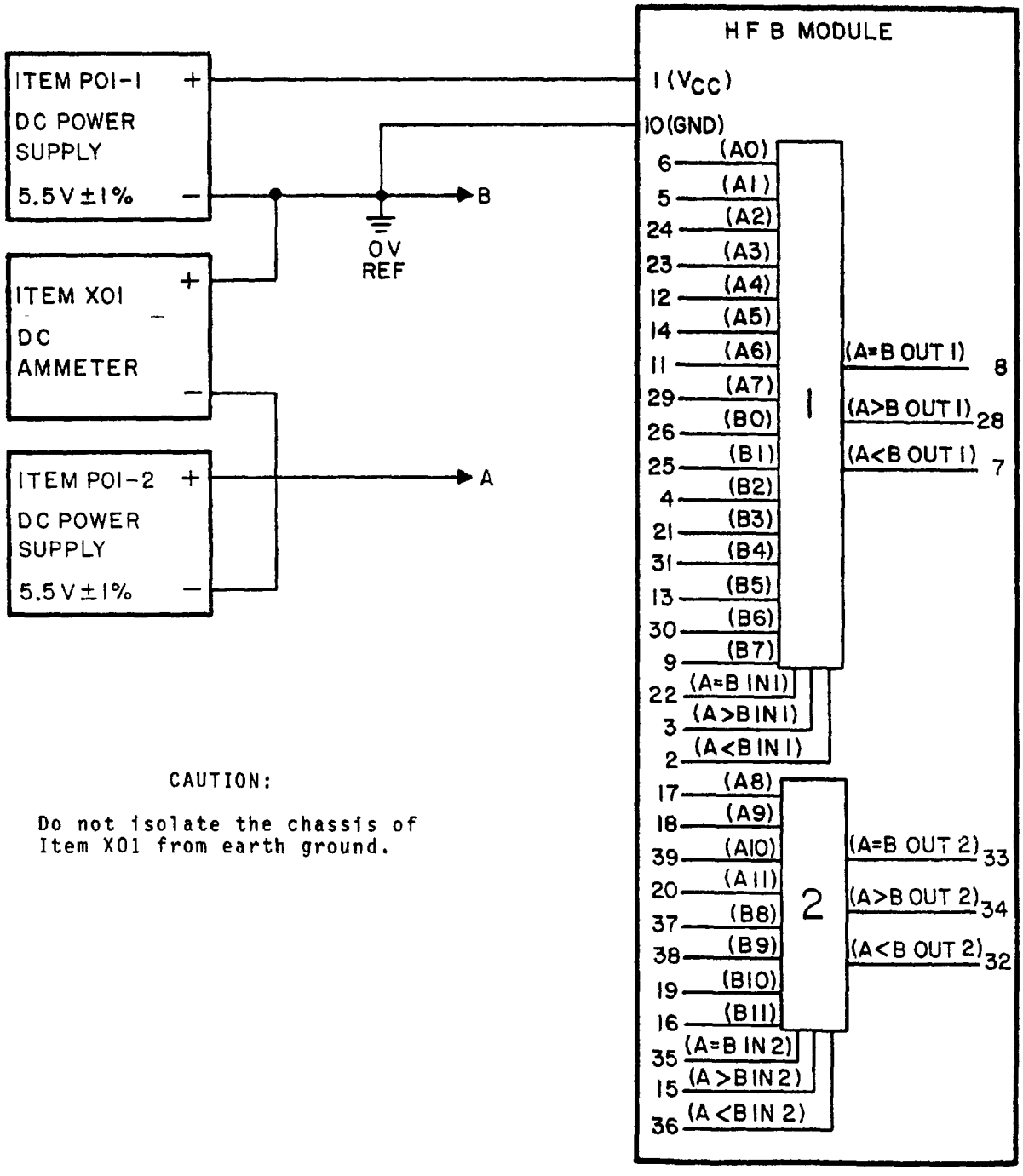
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	A0 Input breakdown current	6	26	0.0	0.3	mA	7
C2	A1 "	5	25	"	"	"	"
C3	A2 "	24	4	"	"	"	"
C4	A3 "	23	21	"	"	"	"
C5	A4 "	12	31	"	"	"	"
C6	A5 "	14	13	"	"	"	"
C7	A6 "	11	30	"	"	"	"
C8	A7 "	29	9	"	"	"	"
C9	A8 "	17	37	"	"	"	"
C10	A9 "	18	38	"	"	"	"
C11	A10 "	39	19	"	"	"	"
C12	A11 "	20	16	"	"	"	"
C13	B0 "	26	6	"	"	"	"
C14	B1 "	25	5	"	"	"	"
C15	B2 "	4	24	"	"	"	"
C16	B3 "	21	23	"	"	"	"
C17	B4 "	31	12	"	"	"	"
C18	B5 "	13	14	"	"	"	"
C19	B6 "	30	11	"	"	"	"
C20	B7 "	9	29	"	"	"	"
C21	B8 "	37	17	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C22	B9 Input breakdown current	38	18	0.0	0.3	mA	7
C23	B10 "	19	39	"	"	"	"
C24	B11 "	16	20	"	"	"	"
C25	A=B IN 1 "	22	26	"	"	"	"
C26	A=B IN 2 "	35	37	"	"	"	"
C27	A<B IN 1 "	2	26	"	0.1	"	"
C28	A>B IN 1 "	3	26	"	"	"	"
C29	A>B IN 2 "	15	37	"	"	"	"
C30	A<B IN 2 "	36	37	"	"	"	"

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CAUTION:

Do not isolate the chassis of Item X01 from earth ground.

NOTE: Power supply P01-2 shall be current limited to 5 mA.

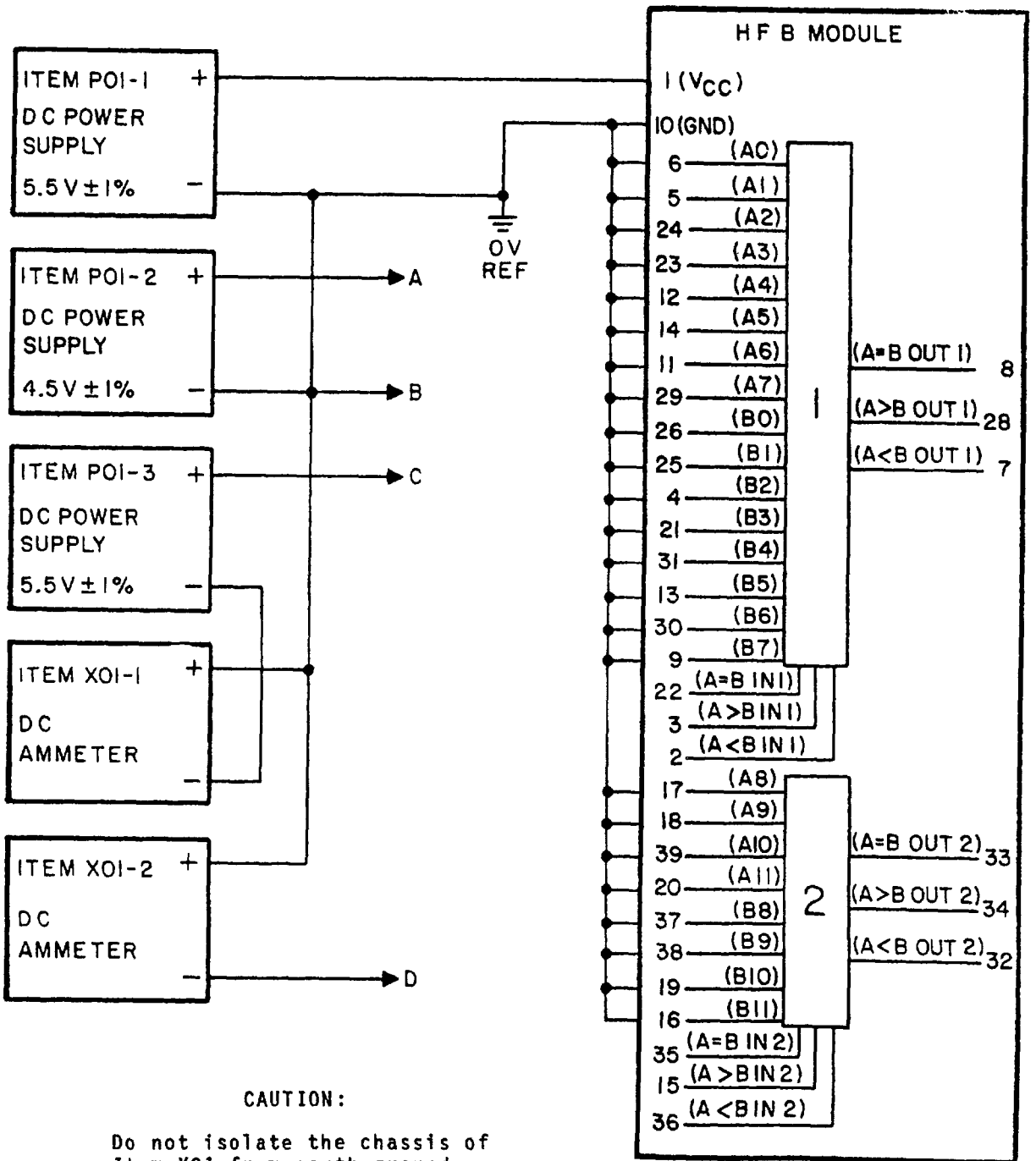
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output leakage current and output short circuit current test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
D1	A=B OUT 1 Output leakage current	22	2,3	8	Open	0	1	mA	8
D2	A>B OUT 1 "	3	2,22	28	"	"	"	"	"
D3	A<B OUT 1 "	2	3,22	7	"	"	"	"	"
D4	A=B OUT 2 "	35	15,36	33	"	"	"	"	"
D5	A>B OUT 2 "	15	35,36	34	"	"	"	"	"
D6	A<B OUT 2 "	36	15,35	32	"	"	"	"	"
E1	A=B OUT 1 Output short circuit current	22	2,3	Open	8	-30	-130	"	"
E2	A>B OUT 1 "	3	2,22	"	28	"	"	"	"
E3	A<B OUT 1 "	2	3,22	"	7	"	"	"	"
E4	A=B OUT 2 "	35	15,36	"	33	"	"	"	"
E5	A>B OUT 2 "	15	35,36	"	34	"	"	"	"
E6	A<B OUT 2 "	36	15,35	"	32	"	"	"	"

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CAUTION:

Do not isolate the chassis of Item X01 from earth ground.

NOTES:
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.
2. For output short circuit, only one output shall be shorted at a time.
3. Power supply P01-3 shall be current limited to 5 mA.

FIGURE 8. Output leakage current and output short circuit current test circuit for table VI.

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TABLE VII. Input clamp diode voltage test.

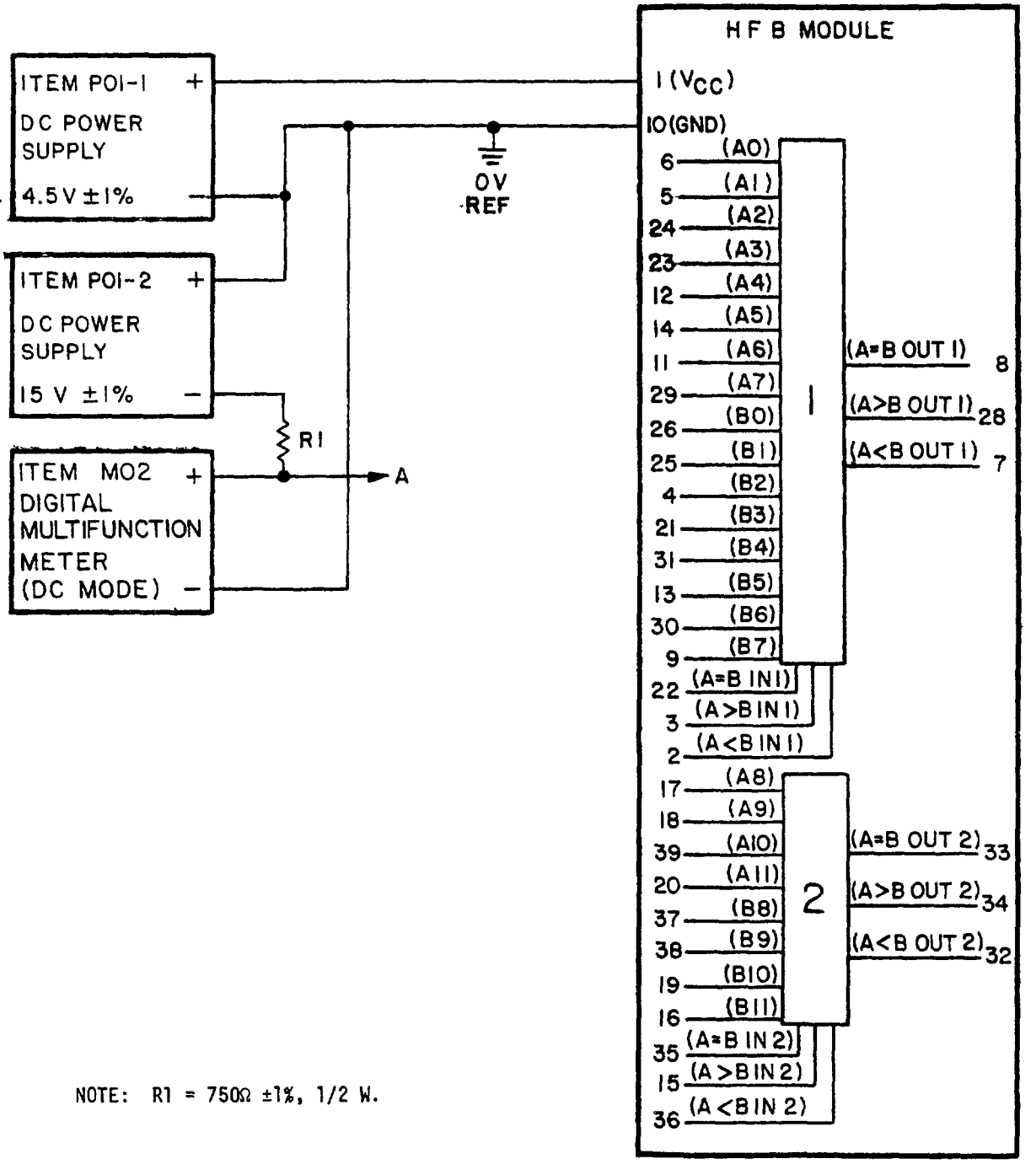
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F1	A0 Input clamp diode voltage	6	-0.4	-1.5	V	9
F2	A1 "	5	"	"	"	"
F3	A2 "	24	"	"	"	"
F4	A3 "	23	"	"	"	"
F5	A4 "	12	"	"	"	"
F6	A5 "	14	"	"	"	"
F7	A6 "	11	"	"	"	"
F8	A7 "	29	"	"	"	"
F9	A8 "	17	"	"	"	"
F10	A9 "	18	"	"	"	"
F11	A10 "	39	"	"	"	"
F12	A11 "	20	"	"	"	"
F13	B0 "	26	"	"	"	"
F14	B1 "	25	"	"	"	"
F15	B2 "	4	"	"	"	"
F16	B3 "	21	"	"	"	"
F17	B4 "	31	"	"	"	"
F18	B5 "	13	"	"	"	"
F19	B6 "	30	"	"	"	"
F20	B7 "	9	"	"	"	"
F21	B8 "	37	"	"	"	"

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TABLE VII. Input clamp diode voltage test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F22	B9	Input clamp diode voltage	38	-0.4	-1.5	V	9
F23	B10	"	19	"	"	"	"
F24	B11	"	16	"	"	"	"
F25	A=B IN 1	"	22	"	"	"	"
F26	A>B IN 1	"	3	"	"	"	"
F27	A<B IN 1	"	2	"	"	"	"
F28	A=B IN 2	"	35	"	"	"	"
F29	A>B IN 2	"	15	"	"	"	"
F30	A<B IN 2	"	36	"	"	"	"

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NOTE: R1 = 750Ω ±1%, 1/2 W.

* FIGURE 9. Input clamp diode voltage test circuit for table VII.

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TABLE VIII. Input current (high-level and low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	A0 Input current, high level	6	Open	1/	0	60	µA	10
G2	A1 "	5	"	"	"	"	"	"
G3	A2 "	24	"	"	"	"	"	"
G4	A3 "	23	"	"	"	"	"	"
G5	A4 "	12	"	"	"	"	"	"
G6	A5 "	14	"	"	"	"	"	"
G7	A6 "	11	"	"	"	"	"	"
G8	A7 "	29	"	"	"	"	"	"
G9	A8 "	17	"	"	"	"	"	"
G10	A9 "	18	"	"	"	"	"	"
G11	A10 "	39	"	"	"	"	"	"
G12	A11 "	20	"	"	"	"	"	"
G13	B0 "	26	"	"	"	"	"	"
G14	B1 "	25	"	"	"	"	"	"
G15	B2 "	4	"	"	"	"	"	"
G16	B3 "	21	"	"	"	"	"	"
G17	B4 "	31	"	"	"	"	"	"
G18	B5 "	13	"	"	"	"	"	"
G19	B6 "	30	"	"	"	"	"	"
G20	B7 "	9	"	"	"	"	"	"
G21	B8 "	37	"	"	"	"	"	"
G22	B9 "	38	"	"	"	"	"	"

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TABLE VIII. Input current (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G23	B10 Input current, high level	19	Open	1/	0	60	µA	10
G24	B11 "	16	"	"	"	"	"	"
G25	A=B IN 1 "	22	"	"	"	"	"	"
G26	A=B IN 2 "	35	"	"	"	"	"	"
G27	A>B IN 1 "	3	"	"	"	20	"	"
G28	A<B IN 1 "	2	"	"	"	"	"	"
G29	A>B IN 2 "	15	"	"	"	"	"	"
G30	A<B IN 2 "	36	"	"	"	"	"	"
H1	A0 Input current, low level	Open	6	Open	-0.10	-1.32	mA	"
H2	A1 "	"	5	"	"	"	"	"
H3	A2 "	"	24	"	"	"	"	"
H4	A3 "	"	23	"	"	"	"	"
H5	A4 "	"	12	"	"	"	"	"
H6	A5 "	"	14	"	"	"	"	"
H7	A6 "	"	11	"	"	"	"	"
H8	A7 "	"	29	"	"	"	"	"
H9	A8 "	"	17	"	"	"	"	"
H10	A9 "	"	18	"	"	"	"	"
H11	A10 "	"	39	"	"	"	"	"
H12	A11 "	"	20	"	"	"	"	"
H13	B0 "	"	26	"	"	"	"	"
H14	B1 "	"	25	"	"	"	"	"

See footnote at end of table.

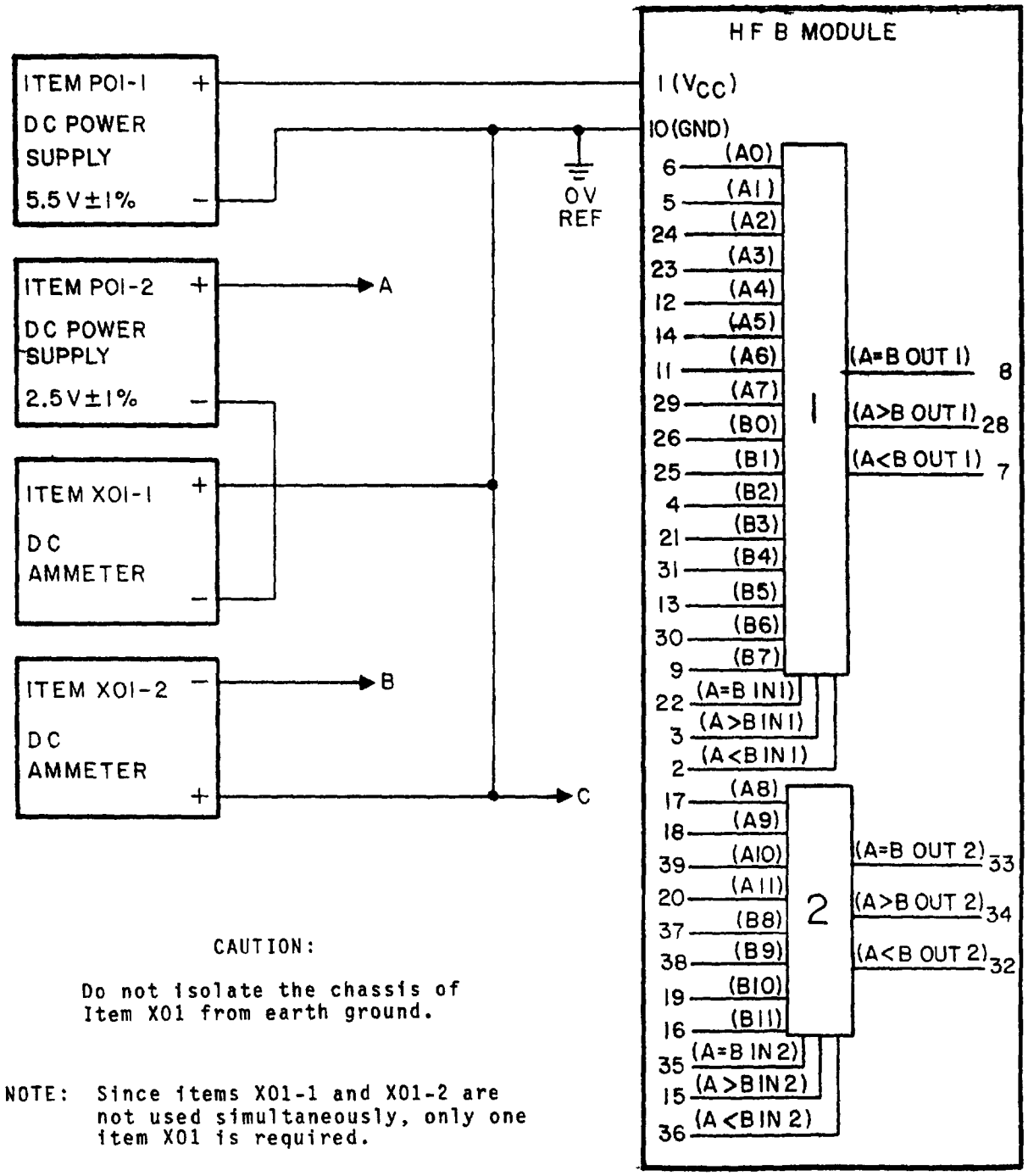
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TABLE VIII. Input current (high-level and low-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H15	B2	Input current, low level	Open	4	Open	-0.10	-1.32	mA	10
H16	B3	"	"	21	"	"	"	"	"
H17	B4	"	"	31	"	"	"	"	"
H18	B5	"	"	13	"	"	"	"	"
H19	B6	"	"	30	"	"	"	"	"
H20	B7	"	"	9	"	"	"	"	"
H21	B8	"	"	37	"	"	"	"	"
H22	B9	"	"	38	"	"	"	"	"
H23	B10	"	"	19	"	"	"	"	"
H24	B11	"	"	16	"	"	"	"	"
H25	A=B IN 1	"	"	22	"	"	"	"	"
H26	A=B IN 2	"	"	35	"	"	"	"	"
H27	A>B IN 1	"	"	3	"	-0.04	-0.44	"	"
H28	A<B IN 1	"	"	2	"	"	"	"	"
H29	A>B IN 2	"	"	15	"	"	"	"	"
H30	A<B IN 2	"	"	36	"	"	"	"	"

1/ All input pins except the one connected to point A should be placed in the C to pin column.

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CAUTION:

Do not isolate the chassis of Item X01 from earth ground.

NOTE: Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required.

FIGURE 10. Input current (high-level and low-level) test circuit for table VIII.

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TABLE IX. Functional compare and output voltage (high-level and low-level) test.

Test condition	Parameter	Connect			Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	A=B	A>B	A<B	Min	Max		
J1	Functional compare	15,35,36,20	16	2/	L	H	L	1/			11
J2	"	15,35,36,16	20	"	L	L	H	"			"
J3	"	15,35,36,16,20	Open	"	H	L	L	"			"
J4	"	15,35,36,	16,20	"	H	L	L	"			"
J5	"	15,35,36,39	19,16,20	"	L	H	L	"			"
J6	"	15,35,36,19	39,16,20	"	L	L	H	"			"
J7	"	15,35,36,19,39	16,20	"	H	L	L	"			"
J8	"	15,35,36	19,39,16,20	"	H	L	L	"			"
J9	"	15,35,36,18	38,16,19,20,39	"	L	H	L	"			"
J10	"	15,35,36,38	18,16,19,20,39	"	L	L	H	"			"
J11	"	15,35,36,18,38	16,19,20,39	"	H	L	L	"			"
J12	"	15,35,36	18,38,16, 19,20,39	"	H	L	L	"			"
J13	"	15,35,36,17	37,16,18,19, 20,38,39	"	L	H	L	"			"
J14	"	15,35,36,37	17,16,18,19, 20,38,39	"	L	L	H	"			"
J15	"	15,35,36,17,37	16,18,19, 20,38,39	"	H	L	L	"			"
J16	"	15,35,36	17,37,16,18, 19,20,38,39	"	H	L	L	"			"
J17	"	35	15,36,16,17,18, 19,20,37,38,39	"	H	L	L	"			"
J18	"	15	35,36,16,17,18, 19,20,37,38,39	"	L	H	L	"			"

See footnotes at end of table.

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TABLE IX. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	A=B	A>B	A<B	Min	Max		
J19	Functional compare	36	15,35,16,17,18, 19,20,37,38,39	2/	L	L	H	1/			11
J20	"	2,3,22,29	9	"	L	H	L	"			"
J21	"	2,3,22,9	29	"	L	L	H	"			"
J22	"	2,3,22,9,29	Open	"	H	L	L	"			"
J23	"	2,3,22	9,29	"	H	L	L	"			"
J24	"	2,3,22,11	30,9,29	"	L	H	L	"			"
J25	"	2,3,22,30	11,9,29	"	L	L	H	"			"
J26	"	2,3,22,11,30	9,29	"	H	L	L	"			"
J27	"	2,3,22	11,30,9,29	"	H	L	L	"			"
J28	"	2,3,22,14	13,9,11,29,30	"	L	H	L	"			"
J29	"	2,3,22,13	14,9,11,29,30	"	L	L	H	"			"
J30	"	2,3,22,13,14	9,11,29,30	"	H	L	L	"			"
J31	"	2,3,22	13,14,9, 11,29,30	"	H	L	L	"			"
J32	"	2,3,22,12	31,9,11,13, 14,29,30,31	"	L	H	L	"			"
J33	"	2,3,22,31	12,9,11,13, 14,29,30	"	L	L	H	"			"
J34	"	2,3,22,12,31	9,11,13, 14,29,30	"	H	L	L	"			"
J35	"	2,3,22	12,31,9,11, 13,14,29,30	"	H	L	L	"			"
J36	"	2,3,22,23	21,9,11,12,13, 14,29,30,31	"	L	H	L	"			"

See footnotes at end of table.

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TABLE IX. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	A=B	A>B	A<B	Min	Max		
J37	Functional compare	2,3,22,21	23,9,11,12,13, 14,29,30,31	2/	L	L	H	1/			11
J38	"	2,3,22,21,23	9,11,12,13,14, 29,30,31	"	H	L	L	"			"
J39	"	2,3,22	21,23,9,11,12, 13,14,29,30,31	"	H	L	L	"			"
J40	"	2,3,22,24	4,9,11,12,13, 14,21,23,29,30, 31	"	L	H	L	"			"
J41	"	2,3,22,4	24,9,11,12,13, 14,21,23,29,30, 31	"	L	L	H	"			"
J42	"	2,3,22,4,24	9,11,12,13,14, 21,23,29,30,31	"	H	L	L	"			"
J43	"	2,3,22	4,24,9,11,12, 13,14,21,23,29, 30,31	"	H	L	L	"			"
J44	"	2,3,22,5	25,4,9,11,12, 13,14,21,23,24, 29,30,31	"	L	H	L	"			"
J45	"	2,3,22,25	5,4,9,11,12,13, 14,21,23,24,29, 30,31	"	L	L	H	"			"
J46	"	2,3,22,5,25	4,9,11,12,13, 14,21,23,24,29, 30,31	"	H	L	L	"			"
J47	"	2,3,22	5,25,4,9,11,12, 13,14,21,23,24, 29,30,31	"	H	L	L	"			"
J48	"	2,3,22,6	26,4,5,9,11,12, 13,14,21,23,24, 29,30,31	"	L	H	L	"			"

See footnotes at end of table.

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TABLE IX. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	A=B	A>B	A<B	Min	Max		
J49	Functional compare	2,3,22,26	6,4,5,9,11,12, 13,14,21,23,24 25,29,30,31	2/	L	L	H	1/			11
J50	"	2,3,22,6,26	4,5,9,11,12,13, 12,14,21,23,24, 25,29,30,31	"	H	L	L	"			"
J51	"	2,3,22	6,26,4,5,9,11, 12,13,14,21,23, 24,25,29,30,31	"	H	L	L	"			"
J52	"	22	2,3,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30, 31	"	H	L	L	"			"
J53	"	3	2,22,4,5,6,9, 11,12,13,14,21, 23,24,25,26,29, 30,31	"	L	H	L	"			"
J54	"	2	3,22,4,5,6,9, 11,12,13,14,21, 23,24,25,26,29, 30,31	"	L	L	H	"			"
K1	A=B Out 1 Output voltage, high level	22	2,3,16,17,18, 19,20,37,38,39	8	H	L	L	2.5	4.5	V	"
K2	A>B Out 1 "	3	2,22	28	L	H	L	"	"	"	"
K3	A<B Out 1 "	2	3,16,17,18,19, 20,22,37,38,39	7	L	L	H	"	"	"	"
K4	A=B Out 2 "	35	4,5,6,9,11,12, 13,14,15,21,23, 24,25,26,29,30, 31,36	33	H	L	L	"	"	"	"
K5	A>B Out 2 "	15	35,36	34	L	H	L	"	"	"	"

See footnotes at end of table.

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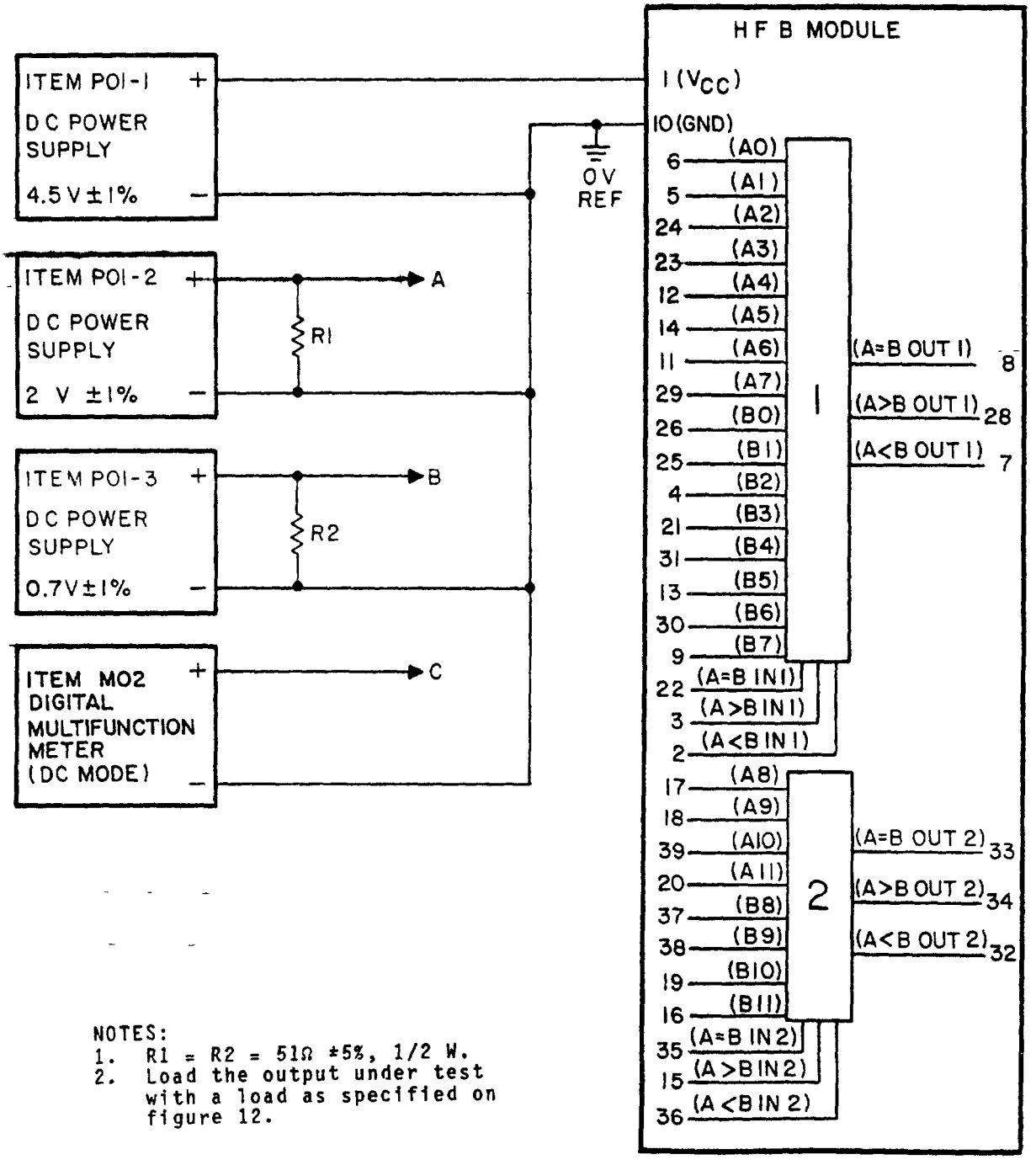
TABLE IX. Functional compare and output voltage (high-level and low-level) test - Continued.

Test condition	Parameter	Connect			Outputs 1/			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	A=B	A>B	A<B	Min	Max		
K6	A<B Out 2 Output voltage, high level	36	4,5,6,9,11,12, 13,14,15,21,24, 25,26,29,30,31, 35	32	L	L	H	2.5	4.5	V	11
L1	A=B Out 1 Output voltage, low level	2	22,3,16,17,18, 19,20,37,38,39	8	L	L	H	0.0	0.4	"	"
L2	A>B Out 1 "	22	3,2	28	H	L	L	"	"	"	"
L3	A<B Out 1 "	2	2,16,17,18,19, 20,22,37,38,39	7	L	H	L	"	"	"	"
L4	A=B Out 2 "	36	35,4,5,6,9,11, 12,13,14,15,21, 23,24,25,26,29, 30,31	33	L	L	H	"	"	"	"
L5	A>B Out 2 "	35	15,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30, 31,36	34	H	L	L	"	"	"	"
L6	A<B Out 2 "	15	36,4,5,6,9,11, 12,13,14,21,23, 24,25,26,29,30, 31,35	32	L	H	L	"	"	"	"

1/ Output logic states:
L = low level voltage $\leq$ 0.4 V.
H = high level voltage $\geq$ 2.4 V.

2/ For test condition J, verification of the output states specified in footnote 1 is obtained by connecting item M02 (point C) to each output pin after input conditions are accomplished.

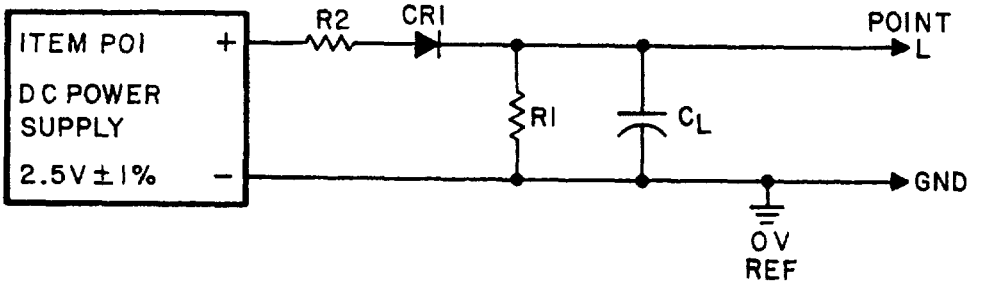
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load the output under test with a load as specified on figure 12.

* FIGURE 11. Functional compare and output voltage (high-level and low-level) test circuit for table IX.

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NOTES:
1. CR1 = 1N4150, 1N3600 or equivalent.
2. R1 = 6.19 kΩ ±1%, 1/8 W.
3. R2 = Select value for a current flow of 4 mA ±1% out of point L with point L held at 0.4 V.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N6, P1-P6, Q1-Q106, R1-R106, T1-T6, and U1-U6.
5. C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions N7-N12, P7-P12, Q107-Q212, and R107-R212.
6. C_L is not required for dc electrical test J, K, and L.
7. Lead length from load circuit to module output pin under test shall be less than 2 inches (50.8 mm).

* FIGURE 12. HFB logic load circuit.

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TABLE X. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
M	Power supply current	15	60	mA	13

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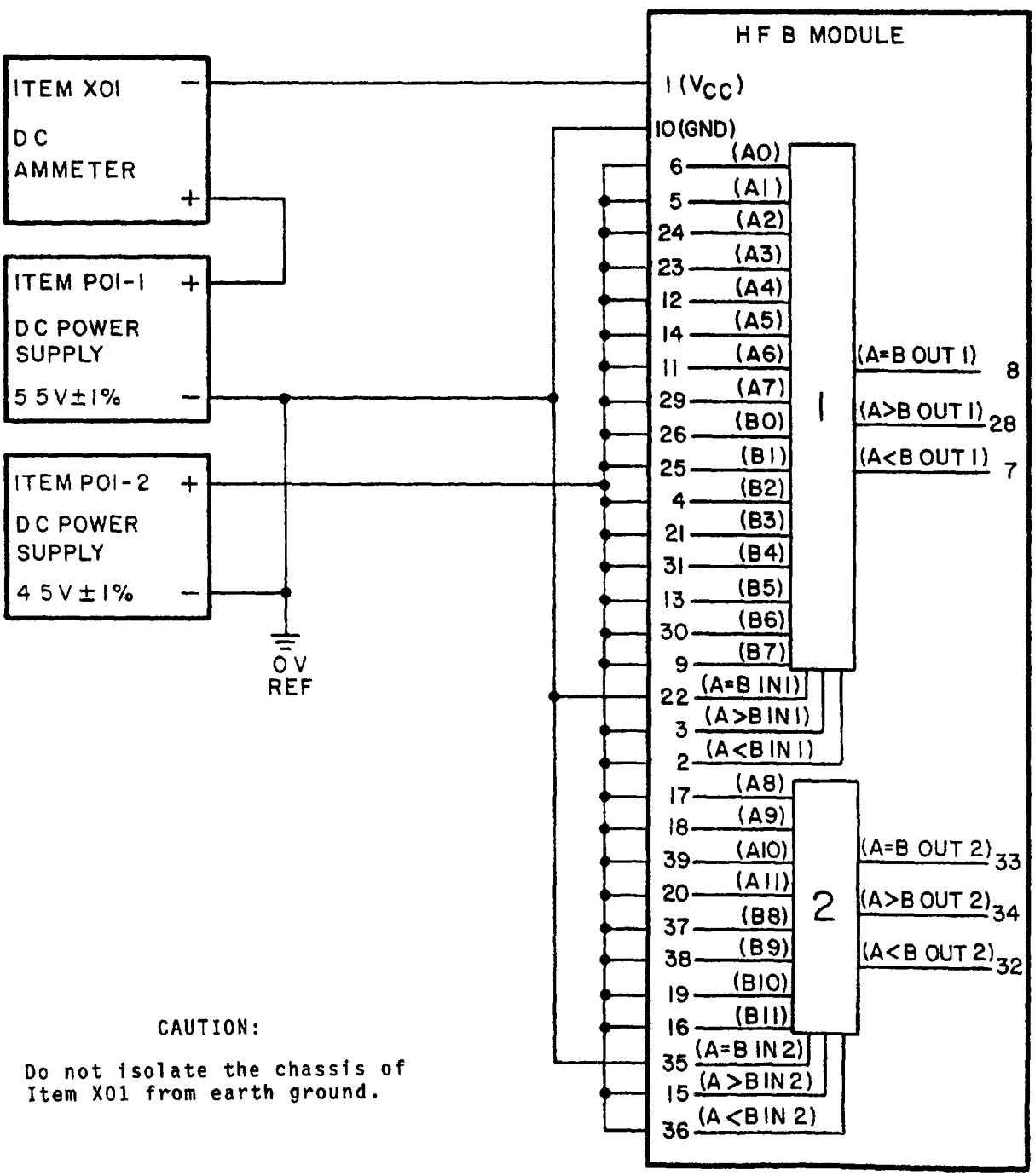


FIGURE 13. Power supply current test circuit for table X.

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TABLE XI. Transition time (t_{TLH} and t_{THL}) test.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
N1	A>B OUT 1	Transition time (t_{TLH}), C_L= 50 pF	Open	3	22	28	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	5	50	ns	14,15
N2	A<B OUT 1	"	"	2	22	7	3	"	"	"	"
N3	A=B OUT 1	"	"	22	3	8	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	"	"	"	"
N4	A>B OUT 2	"	"	15	35	34	16,17,18,19,20,36,37, 38,39	"	"	"	"
N5	A<B OUT 2	"	"	36	35	32	15	"	"	"	"
N6	A=B OUT 2	"	"	35	15	33	16,17,18,19,20,36,37, 38,39	"	"	"	"
N7	A>B OUT 1	Transition time (t_{TLH}), C_L=150 pF	"	3	22	28	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	10	80	"	"
N8	A<B OUT 1	"	"	2	22	7	3	"	"	"	"
N9	A=B OUT 1	"	"	22	3	8	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	"	"	"	"
N10	A>B OUT 2	"	"	15	35	34	16,17,18,19,20,36,37, 38,39	"	"	"	"
N11	A<B OUT 2	"	"	36	35	32	15	"	"	"	"
N12	A=B OUT 2	"	"	35	15	33	16,17,18,19,20,36,37, 38,39	"	"	"	"
P1	A>B OUT 1	Transition time (t_{THL}), C_L= 50 pF	"	3	22	28	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	3	18	"	"
P2	A<B OUT 1	"	"	2	22	7	3	"	"	"	"
P3	A=B OUT 1	"	"	22	3	8	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	"	"	"	"

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TABLE XI. Transition time (t_{TLH} and t_{THL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
P4	A>B OUT 2 Transition time (t_{THL}), C_L= 50 pF	Open	15	35	34	16,17,18,19,20,36,37, 38,39	3	18	ns	14,15
P5	A<B OUT 2 "	"	36	35	32	15	"	"	"	"
P6	A=B OUT 2 "	"	35	15	33	16,17,18,19,20,36,37, 38,39	"	"	"	"
P7	A>B OUT 1 Transition time (t_{THL}), C_L=150 pF	"	3	22	28	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	6	40	"	"
P8	A<B OUT 1 "	"	2	22	7	3	"	"	"	"
P9	A=B OUT 1 "	"	22	3	8	2,4,5,6,9,11,12,13,14, 21,23,24,25,26,29,30, 31	"	"	"	"
P10	A>B OUT 2 "	"	15	35	34	16,17,18,19,20,36,37, 38,39	"	"	"	"
P11	A<B OUT 2 "	"	36	35	32	15	"	"	"	"
P12	A=B OUT 2 "	"	35	15	33	16,17,18,19,20,36,37, 38,39	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q1	A<B to A>B OUT 1 Propagation delay time (t_{PLH}), C_L=50 pF	Open	2	3	28	4,5,6,9,11,12,13,14, 21,22,23,24,25,26,29, 30,31	5	52	ns	14,15
Q2	A=B to A>B OUT 1 "	"	22	3	28	2	"	"	"	"
Q3	A>B to A<B OUT 1 "	"	3	2	7	22	"	"	"	"
Q4	A=B to A<B OUT 1 "	"	22	"	7	3	"	"	"	"
Q5	A=B to A=B OUT 1 "	"	22	"	8	3,4,5,6,9,11,12,13, 14,21,23,24,25,26,29, 30,31	4	48	"	"
Q6	A<B to A>B OUT 2 "	"	36	15	34	16,17,18,19,20,35,37, 38,39	2	26	"	"
Q7	A=B to A>B OUT 2 "	"	35	15	34	36	"	"	"	"
Q8	A>B to A<B OUT 2 "	"	15	36	32	35	"	"	"	"
Q9	A=B to A<B OUT 2 "	"	35	"	32	15	"	"	"	"
Q10	A=B to A=B OUT 2 "	"	35	"	33	15,16,17,18,19,20,37, 38,39	"	24	"	"
Q11	A11>B11, pulse on A, 0 V on B "	"	20	Open	34	16	4	40	"	"
Q12	A10>B10, pulse on A, 0 V on B "	"	39	"	"	16,19,20	"	"	"	"
Q13	A9>B9, pulse on A, 0 V on B "	"	18	"	"	16,19,20,38,39	"	"	"	"
Q14	A8>B8, pulse on A, 0 V on B "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
Q15	A11=B11, pulse on A, 0 V on B "	"	20	"	33	16	"	49	"	"
Q16	A10=B10, pulse on A, 0 V on B "	"	39	"	"	16,19,20	"	49	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q17	A9=B9, pulse on A, 0 V on B; Propagation delay time (t_{PLH}), C_L=50 pF	Open	18	Open	33	16,19,20,38,39	4	49	ns	14,15
Q18	A8=B8, pulse on A, 0 V on B; Propagation delay time (t_{PLH}), C_L=50 pF	Open	17	"	33	16,18,19,20,37,38,39	"	49	"	"
Q19	A11<B11 Propagation delay time (t_{PLH}), C_L=50 pF, pulse on A, 4.5 V on B	16	20	"	32	Open	"	40	"	"
Q20	A10<B10 "	16,19,20	39	"	"	"	"	"	"	"
Q21	A9<B9 "	16,19,20,38,39	18	"	"	"	"	"	"	"
Q22	A8<B8 "	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
Q23	A11=B11 "	16	20	"	33	"	"	49	"	"
Q24	A10=B10 "	16,19,20	39	"	"	"	"	"	"	"
Q25	A9=B9 "	16,19,20,38,39	18	"	"	"	"	"	"	"
Q26	A8=B8 "	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
Q27	A11=B11 Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 4.5 V on A	20	16	"	34	"	"	40	"	"
Q28	A10>B10 "	16,20,39	19	"	"	"	"	"	"	"
Q29	A9>B9 "	16,18,19,20,39	38	"	"	"	"	"	"	"
Q30	A8>B8 "	16,17,18,19,20, 38,39	37	"	"	"	"	"	"	"
Q31	A11=B11 "	20	16	"	33	"	"	49	"	"
Q32	A10=B10 "	16,20,39	19	"	33	"	"	49	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q33	A9=B9 Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 4.5 V on A	16,18,19,20,39	38	Open	33	Open	4	49	ns	14,15
Q34	A8=B8 Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 4.5 V on A	16,17,18,19,20, 38,39	37	"	33	Open	"	49	"	"
Q35	A11<B11 Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 0 V on A	Open	16	"	32	20	"	40	"	"
Q36	A10<B10 "	"	19	"	"	16,20,39	"	"	"	"
Q37	A9<B9 "	"	38	"	"	16,18,19,20,39	"	"	"	"
Q38	A8<B8 "	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
Q39	A11=B11 "	"	16	"	33	20	"	49	"	"
Q40	A10=B10 "	"	19	"	"	16,20,39	"	"	"	"
Q41	A9=B9 "	"	38	"	"	16,18,19,20,39	"	"	"	"
Q42	A8=B8 "	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
Q43	A7>B7 Propagation delay time (t_{PLH}), C_L=50 pF, pulse on A, 0 V on B	"	29	"	28	9	8	80	"	"
Q44	A6>B6 "	"	11	"	"	9,29,30	"	"	"	"
Q45	A5>B5 "	"	14	"	"	9,11,13,29,30	"	"	"	"
Q46	A4>B4 "	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
Q47	A3>B3 "	"	23	"	"	9,11,12,13,14,21,29, 30,31	"	"	"	"
Q48	A2>B2 "	"	24	"	"	4,9,11,12,13,14,21, 23,29,30,31	"	"	"	"
Q49	A1>B1 "	"	5	"	"	4,9,11,12,13,14,21, 23,24,25,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q50	A0>B0	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on A, 0 V on B	Open	6	Open	28	4,5,9,11,12,13,14,21, 23,24,25,26,29,30,31	8	80	ns	14,15
Q51	A7=B7	"	"	29	"	8	9	9	98	"	"
Q52	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
Q53	A5=B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
Q54	A4=B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
Q55	A3=B3	"	"	23	"	"	9,11,12,13,14,21,29, 30,31	"	"	"	"
Q56	A2=B2	"	"	24	"	"	4,9,11,12,13,14,21, 23,29,30,31	"	"	"	"
Q57	A1=B1	"	"	5	"	"	4,9,11,12,13,14,21, 23,24,25,29,30,31	"	"	"	"
Q58	A0=B0	"	"	6	"	"	4,5,9,11,12,13,14, 21,23,24,25,26,29, 30,31	"	"	"	"
Q59	A7>B7	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on A, 4.5 V on B	9	29	"	7	Open	8	80	"	"
Q60	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
Q61	A5<B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
Q62	A4<B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
Q63	A3<B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
Q64	A2<B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
Q65	A1<B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q66	A0>B0	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on A, 4.5 V on B	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	Open	7	Open	8	80	ns	14,15
Q67	A7=B7	"	9	29	"	8	"	9	98	"	"
Q68	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
Q69	A5=B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
Q70	A4=B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
Q71	A3=B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
Q72	A2=B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
Q73	A1=B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"
Q74	A0=B0	"	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	"	"	"	"	"	"	"
Q75	A7>B7	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 4.5 V on A	29	9	"	28	"	8	80	"	"
Q76	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
Q77	A5>B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
Q78	A4>B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
Q79	A3>B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
Q80	A2>B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q81	A1>B1	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 4.5 V on A	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	Open	28	Open	8	80	ns	14,15
Q82	A0>B0	"	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	28	"	8	80	"	"
Q83	A7=B7	"	29	9	"	8	"	9	98	"	"
Q84	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
Q85	A5=B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
Q86	A4=B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
Q87	A3=B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
Q88	A2=B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"
Q89	A1=B1	"	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	"	"	"	"	"	"	"
Q90	A0=B0	"	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	"	"	"	"	"	"
Q91	A7<B7	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 0 V on A	Open	9	"	7	29	8	80	"	"
Q92	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
Q93	A5<B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
Q94	A4<B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
Q95	A3<B3	"	"	21	"	"	9,11,12,13,14,23, 29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q96	A2<B2	Propagation delay time (t_{PLH}), C_L=50 pF, pulse on B, 0 V on A	Open	4	Open	7	9,11,12,13,14,21,23, 24,29,30,31	8	80	ns	14,15
Q97	A1<B1	"	"	25	"	"	4,5,9,11,12,13,14, 21,23,24,29,30,31	"	"	"	"
Q98	A0<B0	"	"	26	"	"	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	"	"	"	"
Q99	A7=B7	"	"	9	"	8	29	9	98	"	"
Q100	A6=B6	"	"	30	"	"	9,11,29	"	"	"	"
Q101	A5=B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
Q102	A4=B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
Q103	A3=B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	"	"	"
Q104	A2=B2	"	"	4	"	"	9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
Q105	A1=B1	"	"	25	"	"	4,5,9,11,12,13,14, 21,23,24,29,30,31	"	"	"	"
Q106	A0=B0	"	"	26	"	"	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	"	"	"	"
Q107	A<B to A>B OUT 1	Propagation delay time (t_{PLH}), C_L=150 pF	"	2	3	28	4,5,6,9,11,12,13,14, 21,22,23,24,25,26, 29,30,31	7	72	"	"
Q108	A=B to A>B OUT 1	"	"	22	3	28	2	"	"	"	"
Q109	A>B to A<B OUT 1	"	"	3	2	7	22	"	"	"	"
Q110	A=B to A<B OUT 1	"	"	22	"	7	3	"	"	"	"
Q111	A=B to A=B OUT 1	"	"	22	"	8	3,4,5,6,9,11,12,13, 14,21,23,24,25,26, 29,30,31	6	68	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q112	A<B to A>B OUT 2	Propagation delay time (t_{PLH}), C_L=150 pF	Open	36	15	34	16,17,18,19,20,35, 37,38,39	3	36	ns	14,15
Q113	A=B to A>B OUT 2	"	"	35	15	34	36	"	"	"	"
Q114	A>B to A<B OUT 2	"	"	15	36	32	35	"	"	"	"
Q115	A=B to A<B OUT 2	"	"	35	"	32	15	"	"	"	"
Q116	A=B to A=B OUT 2	"	"	35	"	33	15,16,17,18,19,20, 37,38,39	"	34	"	"
Q117	A11>B11	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on A, 0 V on B	"	20	Open	34	16	5	50	"	"
Q118	A10>B10	"	"	39	"	"	16,19,20	"	"	"	"
Q119	A9>B9	"	"	18	"	"	16,19,20,38,39	"	"	"	"
Q120	A8>B8	"	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
Q121	A11>B11	"	"	20	"	33	16	"	59	"	"
Q122	A10>B10	"	"	39	"	"	16,19,20	"	"	"	"
Q123	A9=B9	"	"	18	"	"	16,19,20,38,39	"	"	"	"
Q124	A8=B8	"	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
Q125	A11<B11	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on A, 4.5 V on B	16	20	"	32	Open	"	50	"	"
Q126	A10<B10	"	16,19,20	39	"	"	"	"	"	"	"
Q127	A9<B9	"	16,19,20,38,39	18	"	"	"	"	"	"	"
Q128	A8<B8	"	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
Q129	A11=B11	"	16	20	"	33	"	"	59	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q130	A10=B10	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on A, 4.5 V on B	16,19,20	39	Open	33	Open	5	59	ns	14,15
Q131	A9=B9	"	16,19,20,38,39	18	"	"	"	"	"	"	"
Q132	A8=B8	"	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
Q133	A11>B11	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 4.5 V on A	20	16	"	34	"	"	50	"	"
Q134	A10>B10	"	16,20,39	19	"	"	"	"	"	"	"
Q135	A9>B9	"	16,18,19,20,39	38	"	"	"	"	"	"	"
Q136	A8>B8	"	16,17,18,19,20, 38,39	37	"	"	"	"	"	"	"
Q137	A11=B11	"	20	16	"	33	"	"	59	"	"
Q138	A10=B10	"	16,20,39	19	"	"	"	"	"	"	"
Q139	A9=B9	"	16,18,19,20,39	38	"	"	"	"	"	"	"
Q140	A8=B8	"	16,17,18,19,20, 38,39	37	"	"	"	"	"	"	"
Q141	A11<B11	Propagation delay time (t_{PLH}), C_L=150 pF pulse on B, 0 V on A	Open	16	"	32	20	"	50	"	"
Q142	A10<B10	"	"	19	"	"	16,20,39	"	"	"	"
Q143	A9<B9	"	"	38	"	"	16,18,19,20,39	"	"	"	"
Q144	A8<B8	"	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
Q145	A11=B11	"	"	16	"	33	20	"	59	"	"
Q146	A10=B10	"	"	19	"	"	16,20,39	"	"	"	"
Q147	A9=B9	"	"	38	"	"	16,18,19,20,39	"	"	"	"
Q148	A8=B8	"	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q149	A7>B7	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on A, 0 V on B	Open	29	Open	28	9	10	100	ns	14,15
Q150	A6>B6	"	"	11	"	"	9,29,30	"	"	"	"
Q151	A5>B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
Q152	A4>B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
Q153	A3>B3	"	"	23	"	"	9,11,12,13,14,21,29,30,31	"	"	"	"
Q154	A2>B2	"	"	24	"	"	4,9,11,12,13,14,21,23,29,30,31	"	"	"	"
Q155	A1>B1	"	"	5	"	"	4,9,11,12,13,14,21,23,24,25,29,30,31	"	"	"	"
Q156	A0>B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23,24,25,26,29,30,31	"	"	"	"
Q157	A7=B7	"	"	29	"	8	9	11	118	"	"
Q158	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
Q159	A5=B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
Q160	A4=B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
Q161	A3=B3	"	"	23	"	"	9,11,12,13,14,21,29,30,31	"	"	"	"
Q162	A2=B2	"	"	24	"	"	4,9,11,12,13,14,21,23,29,30,31	"	"	"	"
Q163	A1=B1	"	"	5	"	"	4,9,11,12,13,14,21,23,24,25,29,30,31	"	"	"	"
Q164	A0=B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23,24,25,26,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q165	A7<B7	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on A, 4.5 V on B	9	29	Open	7	Open	10	100	ns	14,15
Q166	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
Q167	A5<B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
Q168	A4<B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
Q169	A3<B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
Q170	A2<B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
Q171	A1<B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"
Q172	A0<B0	"	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	"	"	"	"	"	"	"
Q173	A7=B7	"	9	29	"	8	"	11	118	"	"
Q174	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
Q175	A5=B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
Q176	A4=B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
Q177	A3=B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
Q178	A2=B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
Q179	A1=B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q180	A0=B0	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on A, 4.5 V on B	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	Open	8	Open	11	118	ns	14,15
Q181	A7>B7	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 4.5 V on A	29	9	"	28	"	10	100	"	"
Q182	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
Q183	A5>B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
Q184	A4>B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
Q185	A3>B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
Q186	A2>B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"
Q187	A1>B1	"	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	"	28	"	"	"	"	"
Q188	A0>B0	"	4,5,6,9,11,12, 13,14,21,23,24, 29,30,31	26	"	28	"	"	"	"	"
Q189	A7=B7	"	29	9	"	8	"	11	118	"	"
Q190	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
Q191	A5=B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
Q192	A4=B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
Q193	A3=B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
Q194	A2=B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q195	A1=B1	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 4.5 V on A	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	Open	8	Open	11	118	ns	14,15
Q196	A0=B0	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 4.5 V on A	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	8	Open	11	118	"	"
Q197	A7<B7	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 0 V on A	Open	9	"	7	29	10	100	"	"
Q198	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
Q199	A5<B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
Q200	A4<B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
Q201	A3<B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	"	"	"
Q202	A2<B2	"	"	4	"	"	9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
Q203	A1<B1	"	"	25	"	"	4,5,9,11,12,13,14, 21,23,24,29,30,31	"	"	"	"
Q204	A0<B0	"	"	26	"	"	4,5,6,11,12,13,14, 21,23,24,25,29,30,31	"	"	"	"
Q205	A7=B7	"	"	9	"	8	29	11	118	"	"
Q206	A6=B6	"	"	30	"	"	9,11,29	"	"	"	"
Q207	A5=B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
Q208	A4=B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
Q209	A3=B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	"	"	"
Q210	A2=B2	Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 0 V on A	"	4	"	"	9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
Q211	A1=B1 Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 0 V on A	Open	25	Open	8	4,5,9,11,12,13,14, 21,23,24,29,30,31	11	118	ns	14,15
Q212	A0=B0 Propagation delay time (t_{PLH}), C_L=150 pF, pulse on B, 0 V on A	"	26	Open	8	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	11	118	"	"
R1	A<B to A>B OUT 1 Propagation delay time (t_{PHL}), C_L=50 pF	"	2	3	28	4,5,6,9,11,12,13,14, 21,22,23,24,25,26, 29,30,31	4	42	"	"
R2	A=B to A>B OUT 1 "	"	22	3	28	2	"	"	"	"
R3	A>B to A<B OUT 1 "	"	3	2	7	22	"	"	"	"
R4	A=B to A<B OUT 1 "	"	22	"	7	3	"	"	"	"
R5	A=B to A=B OUT 1 "	"	22	"	8	3,4,5,6,9,11,12,13, 14,21,23,24,25,26, 29,30,31	6	60	"	"
R6	A<B to A>B OUT 2 "	"	36	15	34	16,17,18,19,20,35, 37,38,39	2	21	"	"
R7	A=B to A>B OUT 2 "	"	35	15	34	36	"	"	"	"
R8	A>B to A<B OUT 2 "	"	15	36	32	35	"	"	"	"
R9	A=B to A<B OUT 2 "	"	35	"	32	15	"	"	"	"
R10	A=B to A=B OUT 2 "	"	35	"	33	15,16,17,18,19,20, 37,38,39	3	30	"	"
R11	A11>B11 Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 0 V on B	"	20	Open	34	16	"	34	"	"
R12	A10>B10 "	"	39	"	"	16,19,20	"	"	"	"
R13	A9>B9 "	"	18	"	"	16,19,20,38,39	"	"	"	"
R14	A8>B8 "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R15	A11=B11	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 0 V on B	Open	20	Open	33	16	4	49	ns	14,15
R16	A10=B10	"	"	39	"	"	16,19,20	"	"	"	"
R17	A9=B9	"	"	18	"	"	16,19,20,38,39	"	"	"	"
R18	A8=B8	"	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
R19	A11<B11	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 4.5 V on B	16	20	"	32	Open	3	34	"	"
R20	A10<B10	"	16,19,20	39	"	"	"	"	"	"	"
R21	A9<B9	"	16,19,20,38,39	18	"	"	"	"	"	"	"
R22	A8<B8	"	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
R23	A11=B11	"	16	20	"	33	"	4	49	"	"
R24	A10=B10	"	16,19,20	39	"	"	"	"	"	"	"
R25	A9=B9	"	16,19,20,38,39	18	"	"	"	"	"	"	"
R26	A8=B8	"	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
R27	A11>B11	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 4.5 V on A	20	16	"	34	"	3	34	"	"
R28	A10>B10	"	16,20,39	19	"	"	"	"	"	"	"
R29	A9>B9	"	16,18,19,20,39	38	"	"	"	"	"	"	"
R30	A8=B8	"	16,17,18,19,20, 38,39	37	"	"	"	"	"	"	"
R31	A11=B11	"	20	16	"	33	"	4	49	"	"
R32	A10=B10	"	16,20,39	19	"	33	"	4	49	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R33	A9=B9 Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 4.5 V on A	16,18,19,20,39	38	Open	33	Open	4	49	ns	14,15
R34	A8=B8 Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 4.5 V on A	16,17,18,19,20, 38,39	37	"	33	Open	4	49	"	"
R35	A11<B11 Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 0 V on A	Open	16	"	32	20	3	34	"	"
R36	A10<B10 "	"	19	"	"	16,20,39	"	"	"	"
R37	A9<B9 "	"	38	"	"	16,18,19,20,39	"	"	"	"
R38	A8<B8 "	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
R39	A11=B11 "	"	16	"	33	20	4	49	"	"
R40	A10=B10 "	"	19	"	"	16,20,39	"	"	"	"
R41	A9=B9 "	"	38	"	"	16,18,19,20,39	"	"	"	"
R42	A8=B8 "	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
R43	A7>B7 Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 0 V on B	"	29	"	28	9	6	68	"	"
R44	A6>B6 "	"	11	"	"	9,29,30	"	"	"	"
R45	A5>B5 "	"	14	"	"	9,11,13,29,30	"	"	"	"
R46	A4>B4 "	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
R47	A3>B3 "	"	23	"	"	9,11,12,13,14,21,29, 30,31	"	"	"	"
R48	A2>B2 "	"	24	"	28	4,9,11,12,13,14,21, 23,29,30,31	"	"	"	"
R49	A1>B1 "	"	5	"	28	4,9,11,12,13,14,21, 23,24,25,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R50	A0>B0	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 0 V on B	Open	6	Open	28	4,5,9,11,12,13,14,21, 23,24,25,26,29,30,31	6	68	ns	14,15
R51	A7=B7	"	"	29	"	8	9	9	98	"	"
R52	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
R53	A5=B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
R54	A4=B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
R55	A3=B3	"	"	23	"	"	9,11,12,13,14,21,29, 30,31	"	"	"	"
R56	A2=B2	"	"	24	"	"	4,9,11,12,13,14,21, 23,29,30,31	"	"	"	"
R57	A1=B1	"	"	5	"	"	4,9,11,12,13,14,21, 23,24,25,29,30,31	"	"	"	"
R58	A0=B0	"	"	6	"	"	4,5,9,11,12,13,14, 21,23,24,25,26,29, 30,31	"	"	"	"
R59	A7<B7	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 4.5 V on B	9	29	"	7	Open	6	68	"	"
R60	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
R61	A5<B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
R62	A4<B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
R63	A3<B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
R64	A2<B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
R65	A1<B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R66	A0>B0	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on A, 4.5 V on B	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	Open	7	Open	6	68	ns	14,15
R67	A7=B7	"	9	29	"	8	"	9	98	"	"
R68	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
R69	A5=B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
R70	A4=B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
R71	A3=B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
R72	A2=B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
R73	A1=B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"
R74	A0=B0	"	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	"	"	"	"	"	"	"
R75	A7>B7	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 4.5 V on A	29	9	"	28	"	6	68	"	"
R76	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
R77	A5>B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
R78	A4>B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
R79	A3>B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
R80	A2>B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R81	A1>B1	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 4.5 V on A	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	Open	28	Open	6	68	ns	14,15
R82	A0>B0	"	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	28	"	6	68	"	"
R83	A7=B7	"	29	9	"	8	"	9	98	"	"
R84	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
R85	A5=B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
R86	A4=B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
R87	A3=B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
R88	A2=B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"
R89	A1=B1	"	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	"	"	"	"	"	"	"
R90	A0=B0	"	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	"	"	"	"	"	"
R91	A7<B7	Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 0 V on A	Open	9	"	7	29	6	68	"	"
R92	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
R93	A5<B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
R94	A4<B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
R95	A3<B3	"	"	21	"	"	9,11,12,13,14,23, 29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R96	A2<B2 Propagation delay time (t_{PHL}), C_L=50 pF, pulse on B, 0 V on A	Open	4	Open	7	9,11,12,13,14,21,23, 24,29,30,31	6	68	ns	14,15
R97	A1<B1 "	"	25	"	"	4,5,9,11,12,13,14, 21,23,24,29,30,31	"	"	"	"
R98	A0<B0 "	"	26	"	"	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	"	"	"	"
R99	A7=B7 "	"	9	"	8	29	9	98	"	"
R100	A6=B6 "	"	30	"	"	9,11,29	"	"	"	"
R101	A5=B5 "	"	13	"	"	9,11,14,29,30	"	"	"	"
R102	A4=B4 "	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
R103	A3=B3 "	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	"	"	"
R104	A2=B2 "	"	4	"	"	9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
R105	A1=B1 "	"	25	"	"	4,5,9,11,12,13,14, 21,23,24,29,30,31	"	"	"	"
R106	A0=B0 "	"	26	"	"	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	"	"	"	"
R107	A<B to A>B OUT 1 Propagation delay time (t_{PHL}), C_L=150 pF	"	2	3	28	4,5,6,9,11,12,13,14, 21,22,23,24,25,26, 29,30,31	6	62	"	"
R108	A=B to A>B OUT 1 "	"	22	3	28	2	"	"	"	"
R109	A>B to A<B OUT 1 "	"	3	2	7	22	"	"	"	"
R110	A=B to A<B OUT 1 "	"	22	"	7	3	"	"	"	"
R111	A=B to A=B OUT 1 "	"	22	"	8	3,4,5,6,9,11,12,13, 14,21,23,24,25,26, 29,30,31	8	80	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R112	A<B to A>B OUT 2 Propagation delay time (t_{PHL}), C_L=150 pF	Open	36	15	34	16,17,18,19,20,35, 37,38,39	3	31	ns	14,15
R113	A=B to A>B OUT 2 "	"	35	15	34	36	"	"	"	"
R114	A>B to A<B OUT 2 "	"	15	36	32	35	"	"	"	"
R115	A=B to A<B OUT 2 "	"	35	"	32	15	"	"	"	"
R116	A=B to A=B OUT 2 "	"	35	"	33	15,16,17,18,19,20, 37,38,39	4	40	"	"
R117	A11>B11 Propagation delay time (t_{PHL}), C_L=150 pF, pulse on A, 0 V on B	"	20	Open	34	16	"	44	"	"
R118	A10>B10 "	"	39	"	"	16,19,20	"	"	"	"
R119	A9>B9 "	"	18	"	"	16,19,20,38,39	"	"	"	"
R120	A8>B8 "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
R121	A11=B11 "	"	20	"	33	16	5	59	"	"
R122	A10=B10 "	"	39	"	"	16,19,20	"	"	"	"
R123	A9=B9 "	"	18	"	"	16,19,20,38,39	"	"	"	"
R124	A8=B8 "	"	17	"	"	16,18,19,20,37,38,39	"	"	"	"
R125	A11<B11 Propagation delay time (t_{PHL}), C_L=150 pF, pulse on A, 4.5 V on B	16	20	"	32	Open	4	44	"	"
R126	A10<B10 "	16,19,20	39	"	"	"	"	"	"	"
R127	A9<B9 "	16,19,20,38,39	18	"	"	"	"	"	"	"
R128	A8<B8 "	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
R129	A11=B11 "	16	20	"	33	"	5	59	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R130	A10=B10	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on A, 4.5 V on B	16,19,20	39	Open	33	Open	5	59	ns	14,15
R131	A9=B9	"	16,19,20,38,39	18	"	"	"	"	"	"	"
R132	A8=B8	"	16,18,19,20,37, 38,39	17	"	"	"	"	"	"	"
R133	A11>B11	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 4.5 V on A	20	16	"	34	"	4	44	"	"
R134	A10>B10	"	16,20,39	19	"	"	"	"	"	"	"
R135	A9>B9	"	16,18,19,20,39	38	"	"	"	"	"	"	"
R136	A8>B8	"	16,17,18,19,20, 38,39	37	"	"	"	"	"	"	"
R137	A11=B11	"	20	16	"	33	"	5	59	"	"
R138	A10=B10	"	16,20,39	19	"	"	"	"	"	"	"
R139	A9=B9	"	16,18,19,20,39	38	"	"	"	"	"	"	"
R140	A8=B8	"	16,17,18,19,20, 38,39	37	"	"	"	"	"	"	"
R141	A11<B11	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 0 V on A	Open	16	"	32	20	4	44	"	"
R142	A10<B10	"	"	19	"	"	16,20,39	"	"	"	"
R143	A9<B9	"	"	38	"	"	16,18,19,20,39	"	"	"	"
R144	A8<B8	"	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"
R145	A11=B11	"	"	16	"	33	20	5	59	"	"
R146	A10=B10	"	"	19	"	"	16,20,39	"	"	"	"
R147	A9=B9	"	"	38	"	"	16,18,19,20,39	"	"	"	"
R148	A8=B8	"	"	37	"	"	16,17,18,19,20,38,39	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R149	A7>B7	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on A, 0 V on B	Open	29	Open	28	9	8	88	ns	14,15
R150	A6>B6	"	"	11	"	"	9,29,30	"	"	"	"
R151	A5>B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
R152	A4>B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
R153	A3>B3	"	"	23	"	"	9,11,12,13,14,21,29,30,31	"	"	"	"
R154	A2>B2	"	"	24	"	"	4,9,11,12,13,14,21,23,29,30,31	"	"	"	"
R155	A1>B1	"	"	5	"	"	4,9,11,12,13,14,21,23,24,25,29,30,31	"	"	"	"
R156	A0>B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23,24,25,26,29,30,31	"	"	"	"
R157	A7=B7	"	"	29	"	8	9	11	118	"	"
R158	A6=B6	"	"	11	"	"	9,29,30	"	"	"	"
R159	A5=B5	"	"	14	"	"	9,11,13,29,30	"	"	"	"
R160	A4=B4	"	"	12	"	"	9,11,13,14,29,30,31	"	"	"	"
R161	A3=B3	"	"	23	"	"	9,11,12,13,14,21,29,30,31	"	"	"	"
R162	A2=B2	"	"	24	"	"	4,9,11,12,13,14,21,23,29,30,31	"	"	"	"
R163	A1=B1	"	"	5	"	"	4,9,11,12,13,14,21,23,24,25,29,30,31	"	"	"	"
R164	A0=B0	"	"	6	"	"	4,5,9,11,12,13,14,21,23,24,25,26,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R165	A7<B7	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on A, 4.5 V on B	9	29	Open	7	Open	8	88	ns	14,15
R166	A6<B6	"	9,29,30	11	"	"	"	"	"	"	"
R167	A5<B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
R168	A4<B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
R169	A3<B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
R170	A2<B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
R171	A1<B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"
R172	A0<B0	"	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	"	"	"	"	"	"	"
R173	A7=B7	"	9	29	"	8	"	11	118	"	"
R174	A6=B6	"	9,29,30	11	"	"	"	"	"	"	"
R175	A5=B5	"	9,11,13,29,30	14	"	"	"	"	"	"	"
R176	A4=B4	"	9,11,13,14,29, 30,31	12	"	"	"	"	"	"	"
R177	A3=B3	"	9,11,12,13,14, 21,29,30,31	23	"	"	"	"	"	"	"
R178	A2=B2	"	4,9,11,12,13,14, 21,23,29,30,31	24	"	"	"	"	"	"	"
R179	A1=B1	"	4,9,11,12,13,14, 21,23,24,25,29, 30,31	5	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R180	A0=B0	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on A, 4.5 V on B	4,5,9,11,12,13, 14,21,23,24,25, 26,29,30,31	6	Open	8	Open	11	118	ns	14,15
R181	A7>B7	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 4.5 V on A	29	9	"	28	"	8	88	"	"
R182	A6>B6	"	9,11,29	30	"	"	"	"	"	"	"
R183	A5>B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
R184	A4>B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
R185	A3>B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
R186	A2>B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"
R187	A1>B1	"	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	"	"	"	"	"	"	"
R188	A0>B0	"	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	"	"	"	"	"	"
R189	A7=B7	"	29	9	"	8	"	11	118	"	"
R190	A6=B6	"	9,11,29	30	"	"	"	"	"	"	"
R191	A5=B5	"	9,11,14,29,30	13	"	"	"	"	"	"	"
R192	A4=B4	"	9,11,12,13,14, 29,30	31	"	"	"	"	"	"	"
R193	A3=B3	"	9,11,12,13,14, 23,29,30,31	21	"	"	"	"	"	"	"
R194	A2=B2	"	9,11,12,13,14, 21,23,24,29,30, 31	4	"	"	"	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

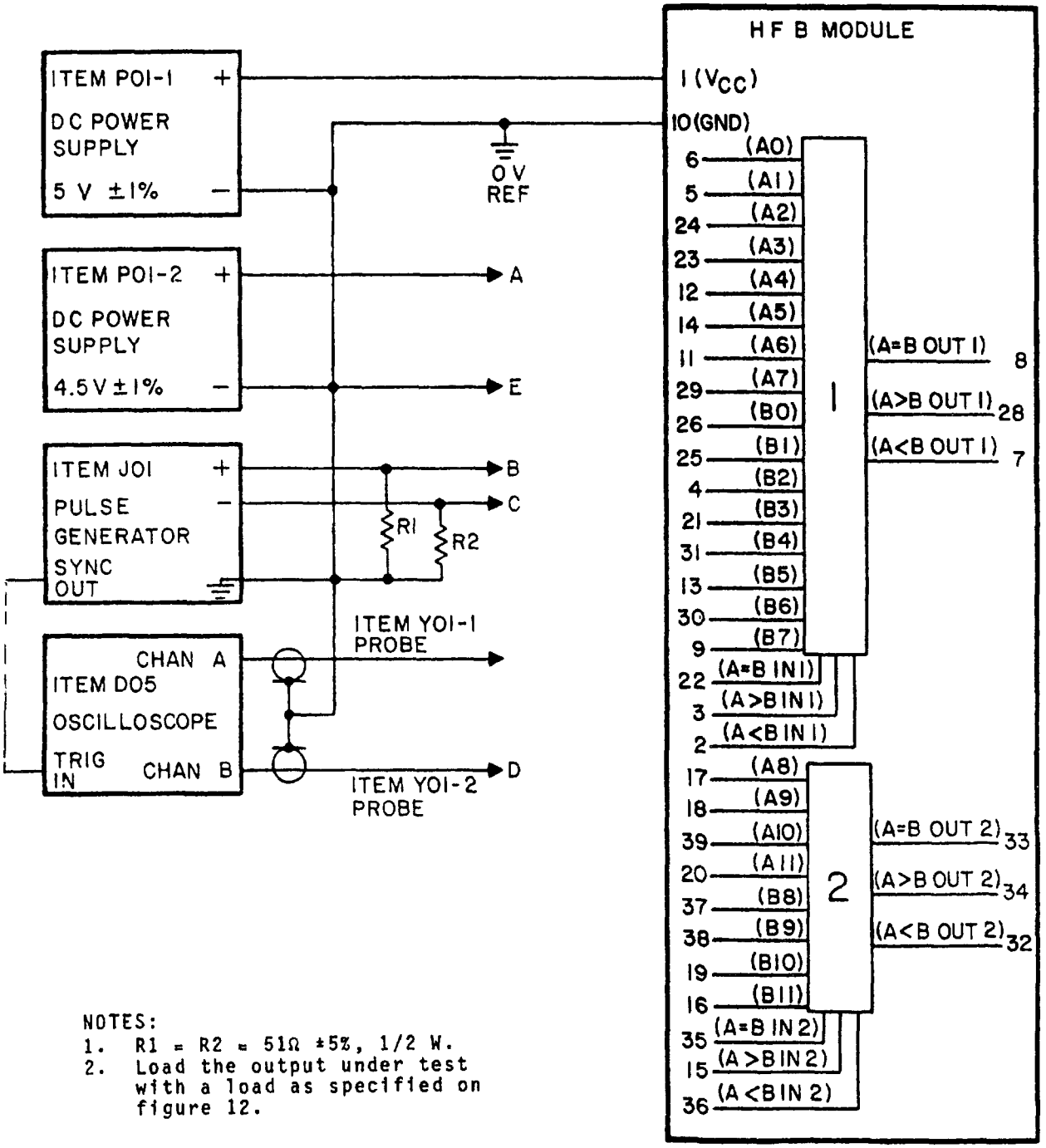
Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R195	A1=B1	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 4.5 V on A	4,5,9,11,12,13, 14,21,23,24,29, 30,31	25	Open	8	Open	11	118	ns	14,15
R196	A0=B0	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 4.5 V on A	4,5,6,9,11,12, 13,14,21,23,24, 25,29,30,31	26	"	8	Open	11	118	"	"
R197	A7<B7	Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 0 V on A	Open	9	"	7	29	8	88	"	"
R198	A6<B6	"	"	30	"	"	9,11,29	"	"	"	"
R199	A5<B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
R200	A4<B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
R201	A3<B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	"	"	"
R202	A2<B2	"	"	4	"	"	9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"
R203	A1<B1	"	"	25	"	"	4,5,9,11,12,13,14, 21,23,24,29,30,31	"	"	"	"
R204	A0<B0	"	"	26	"	"	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	"	"	"	"
R205	A7=B7	"	"	9	"	8	29	11	118	"	"
R206	A6=B6	"	"	30	"	"	9,11,29	"	"	"	"
R207	A5=B5	"	"	13	"	"	9,11,14,29,30	"	"	"	"
R208	A4=B4	"	"	31	"	"	9,11,12,13,14,29,30	"	"	"	"
R209	A3=B3	"	"	21	"	"	9,11,12,13,14,23,29, 30,31	"	"	"	"
R210	A2=B2	"	"	4	"	"	9,11,12,13,14,21,23, 24,29,30,31	"	"	"	"

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TABLE XII. Propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
R211	A1=B1 Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 0 V on A	Open	25	Open	8	4,5,9,11,12,13,14, 21,23,24,29,30,31	11	118	ns	14,15
R212	A0=B0 Propagation delay time (t_{PHL}), C_L=150 pF, pulse on B, 0 V on A	Open	26	Open	8	4,5,6,9,11,12,13,14, 21,23,24,25,29,30,31	11	118	ns	14,15

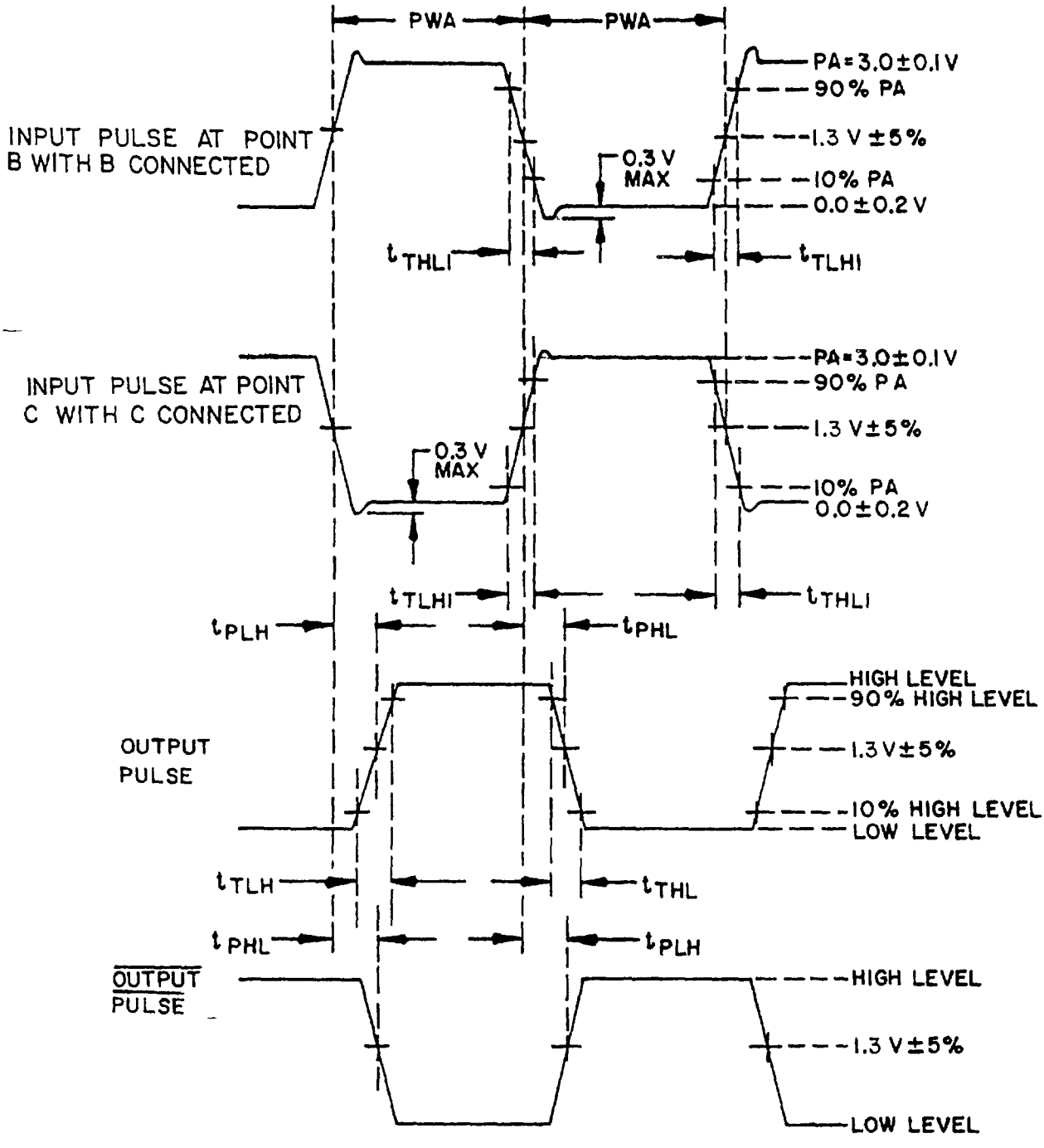
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load the output under test with a load as specified on figure 12.

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XI and XII.

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NOTES:
1. Input pulse parameters:
   PWA = 500 ±50 ns.
   $t_{TLH1} = t_{THL1}$ = 6 ±2 ns.
2. Before making measurements, the output levels shall be verified to be 0.4 V maximum for the low level and 2.5 V minimum for the high level.

FIGURE 15. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XI and XII.

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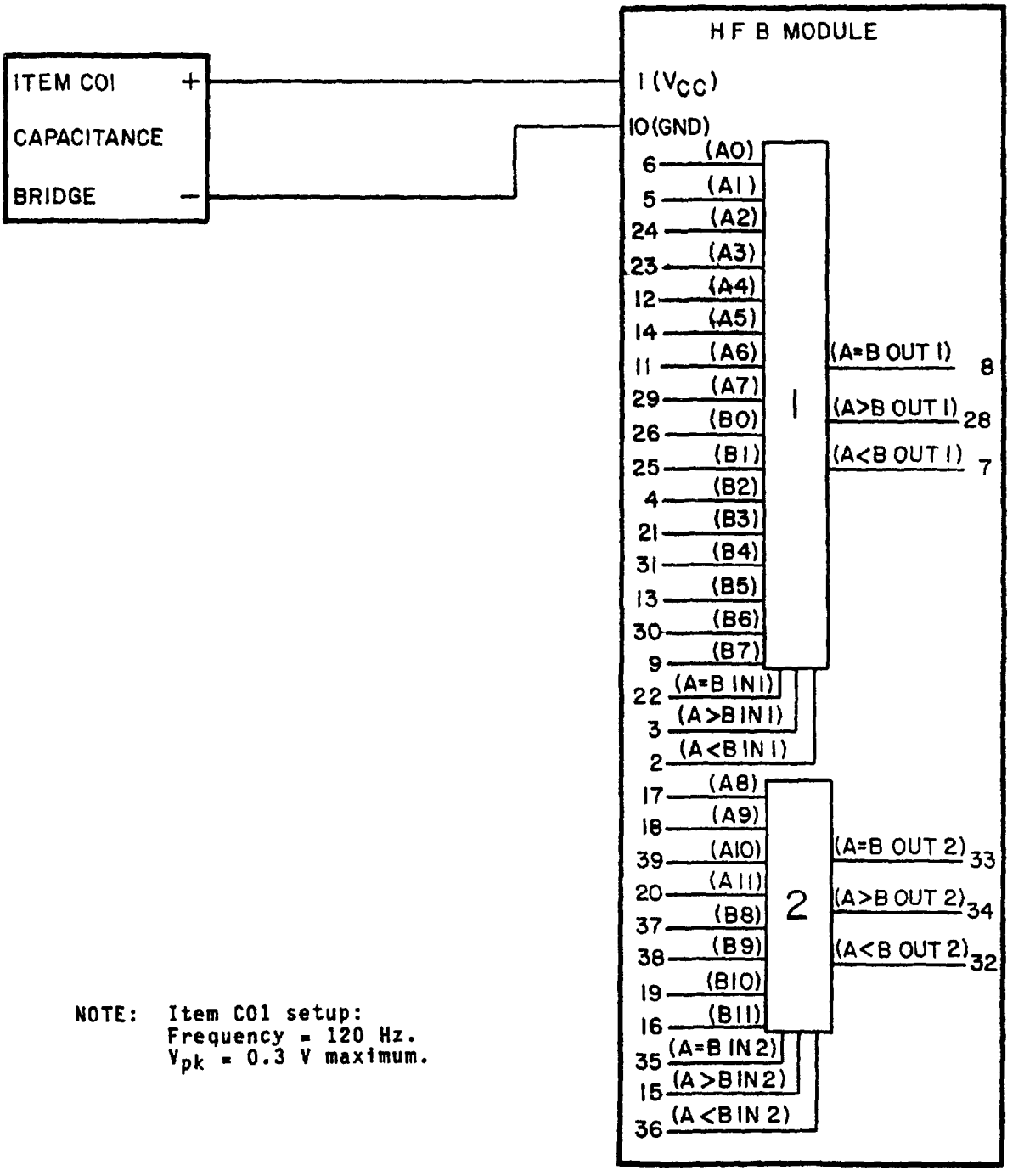
TABLE XIII. Power supply filter capacitance test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply filter capacitance	1.6	3.0	µF	16

TABLE XIV. Output crosstalk (high-level and low-level) test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
T1	A>B OUT 1 Output crosstalk, high level	3	2,15,22	35	36	28	0	400	mV	17,18
T2	A<B OUT 1 "	2	3,22,36	35	15	7	"	"	"	"
T3	A=B OUT 1 "	22	2,3,35	36	15	8	"	"	"	"
T4	A>B OUT 2 "	15	3,35,36	22	2	34	"	"	"	"
T5	A<B OUT 2 "	36	2,15,35	22	3	32	"	"	"	"
T6	A=B OUT 2 "	35	15,22,36	2	3	33	"	"	"	"
U1	A>B OUT 1 Output crosstalk, low level	22	2,3,15	35	36	28	"	500	"	"
U2	A<B OUT 1 "	22	2,3,36	35	15	7	"	"	"	"
U3	A=B OUT 1 "	2	3,22,35	36	15	8	"	"	"	"
U4	A>B OUT 2 "	35	15,3,36	22	2	34	"	"	"	"
U5	A<B OUT 2 "	35	2,15,36	22	3	32	"	"	"	"
U6	A=B OUT 2 "	36	15,22,35	2	3	33	"	"	"	"

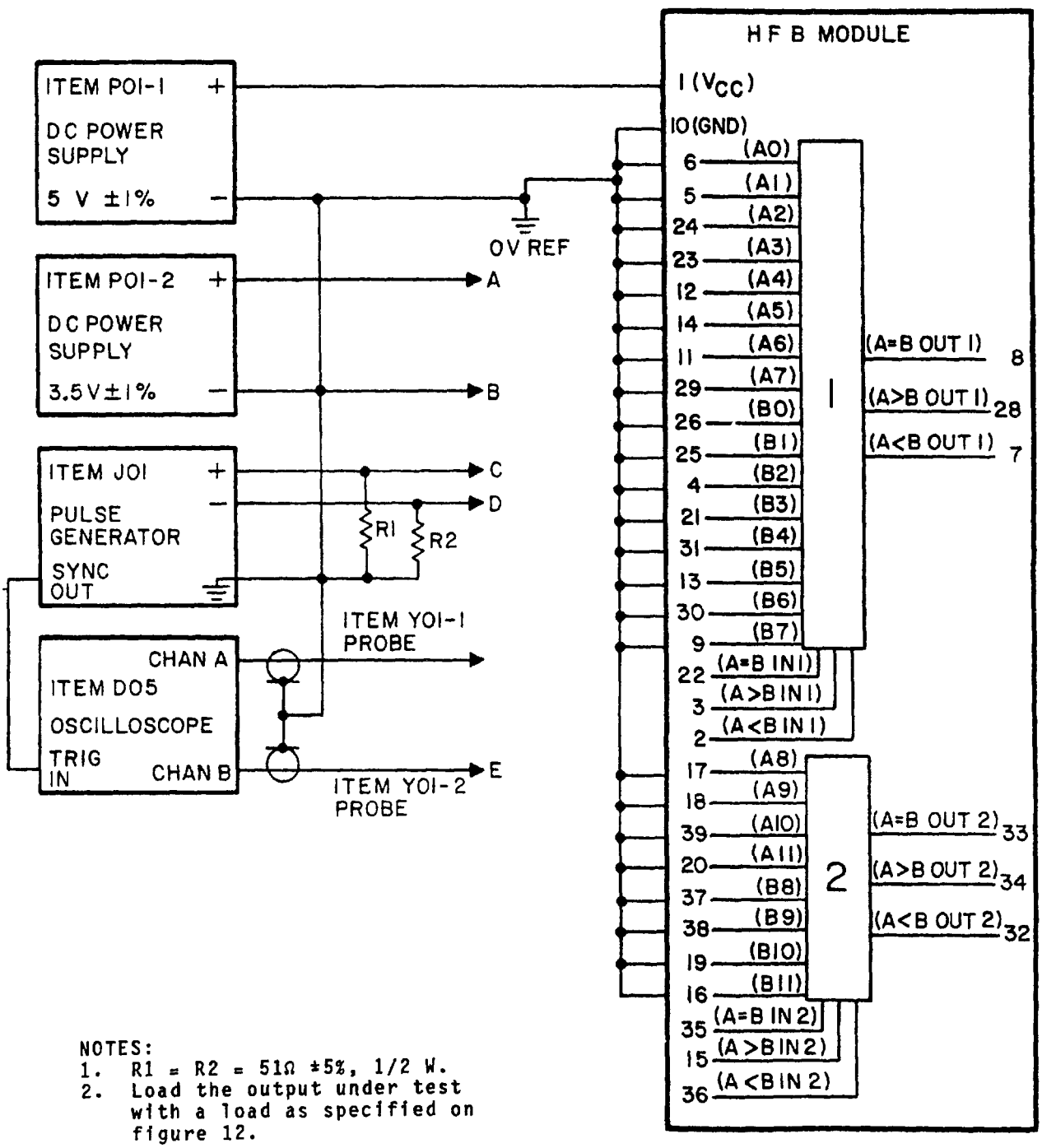
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NOTE: Item C01 setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V maximum.

FIGURE 16. Power supply filter capacitance test circuit for table XIII.

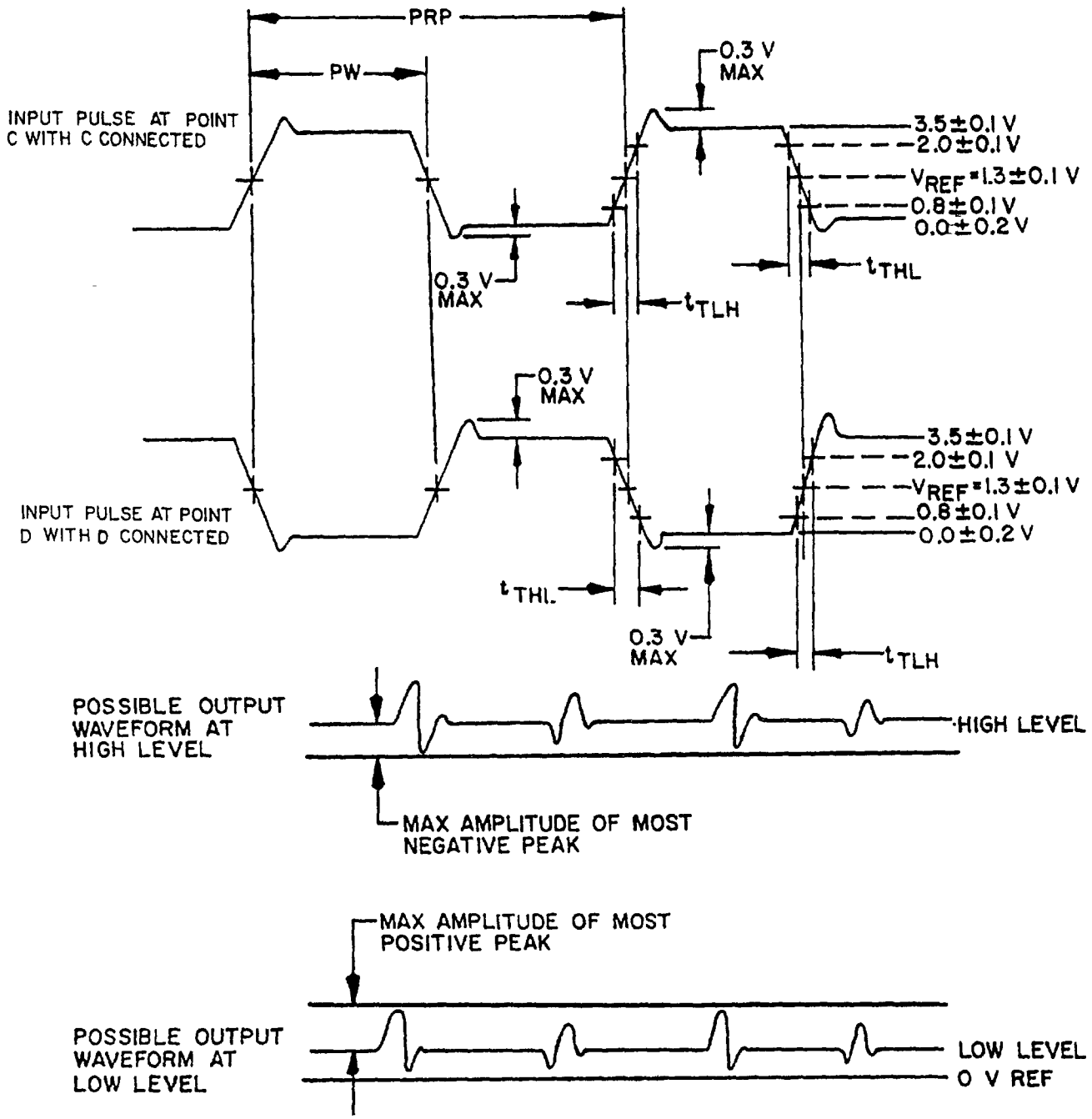
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load the output under test with a load as specified on figure 12.

IGURE 17. Output crosstalk (high-level and low-level) test circuit for table XIV.

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NOTES:
1. Input pulse parameters:
   a. $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   b. PW = 500 ±50 ns.
   c. PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins, probe Y01-1 and Y01-2 to input and output pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins shall be less than 2 inches (50.8 mm).

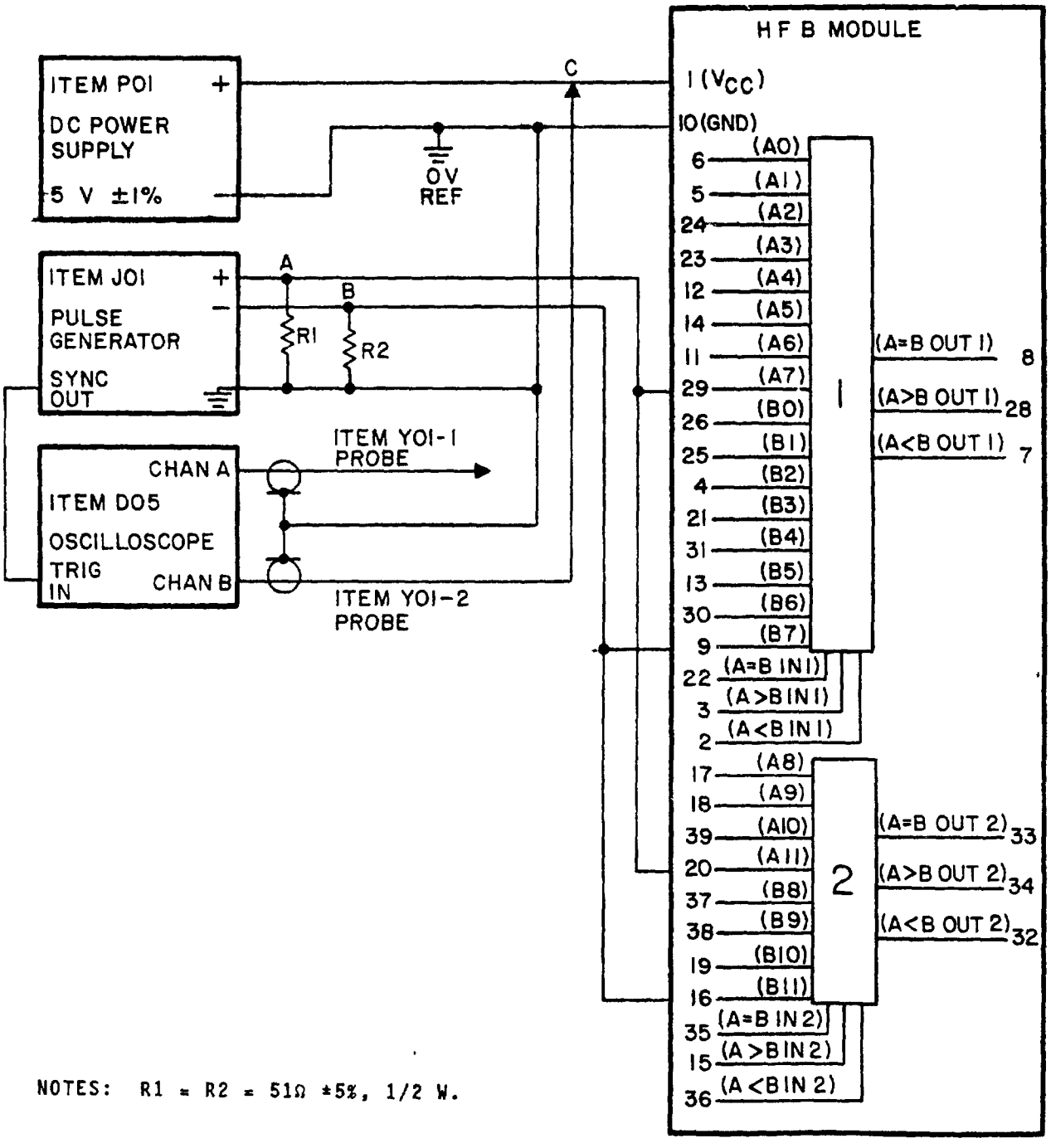
FIGURE 18. Output crosstalk (high-level and low-level) test circuit waveforms for table XIV.

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TABLE XV. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
V	Power supply transient amplitude	0	500	mV	19,20

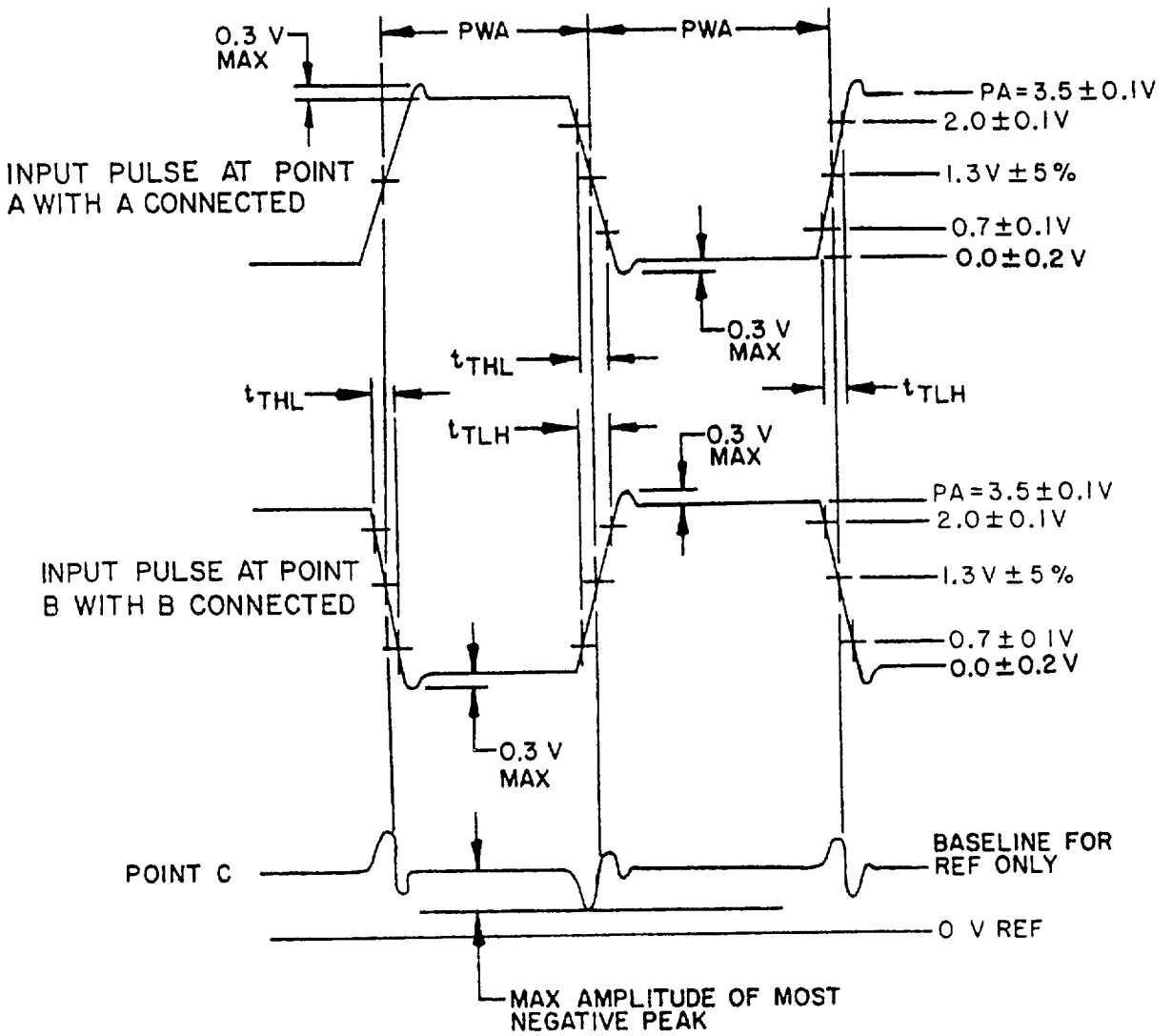
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NOTES: R1 = R2 = 51Ω ±5%, 1/2 W.

FIGURE 19. Power supply transient amplitude test circuit for table XV.

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NOTE Input pulse parameters:

PWA = 500 ±50 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at each input pin).

FIGURE 20. Power supply transient amplitude test circuit waveforms for table XV.

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TABLE XVI. Input capacitance test.

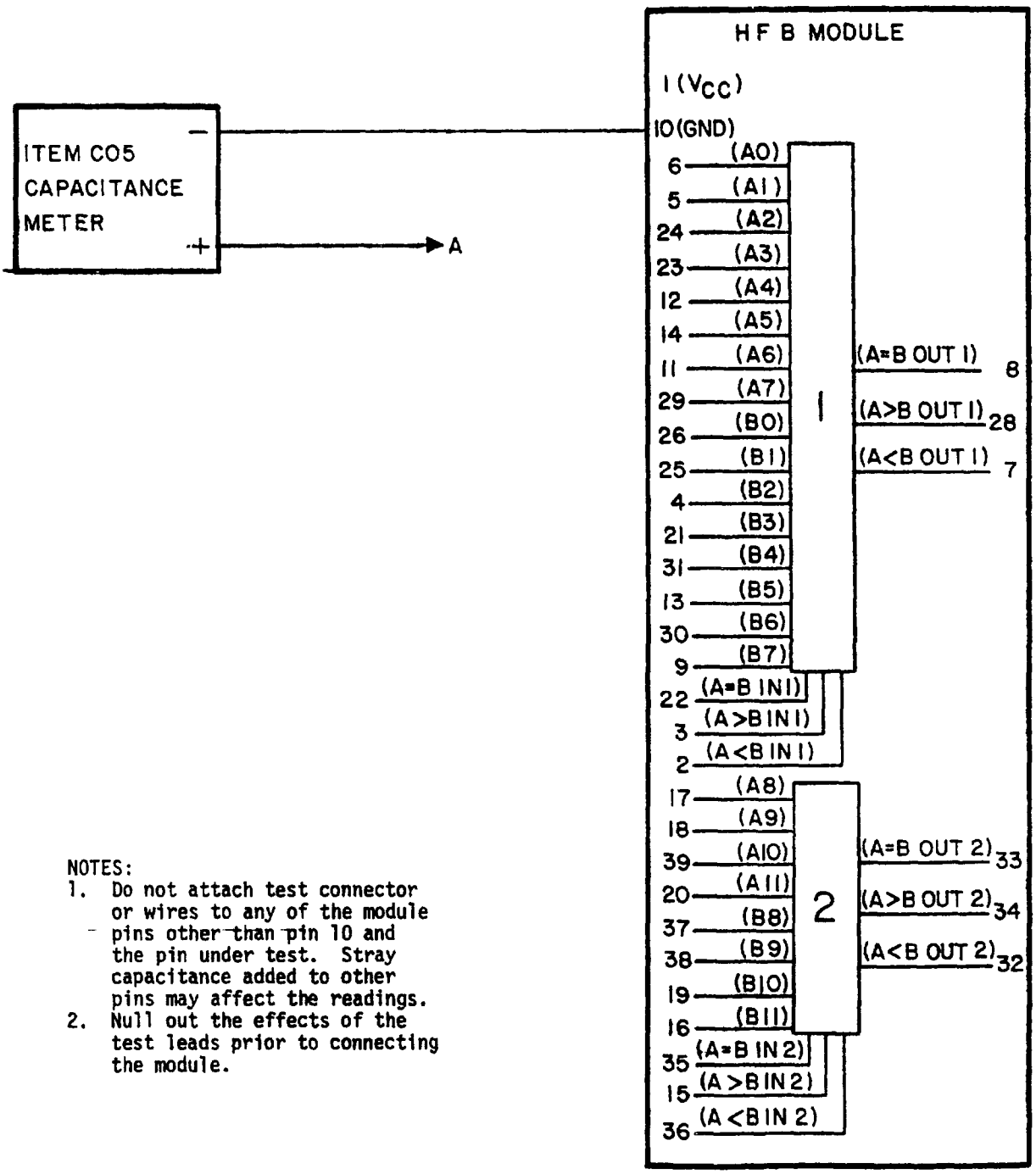
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
W1	A>B IN 1 Input capacitance	3	2	15	pF	21
W2	A<B IN 1 "	2	"	"	"	"
W3	A>B IN 2 "	15	"	"	"	"
W4	A<B IN 2 "	36	"	"	"	"
W5	A=B IN 1 "	22	"	"	"	"
W6	A=B IN 2 "	35	"	"	"	"
W7	A0 "	6	"	"	"	"
W8	A1 "	5	"	"	"	"
W9	A2 "	24	"	"	"	"
W10	A3 "	23	"	"	"	"
W11	A4 "	12	"	"	"	"
W12	A5 "	14	"	"	"	"
W13	A6 "	11	"	"	"	"
W14	A7 "	29	"	"	"	"
W15	A8 "	17	"	"	"	"
W16	A9 "	18	"	"	"	"
W17	A10 "	39	"	"	"	"
W18	A11 "	20	"	"	"	"
W19	B0 "	26	"	"	"	"
W20	B1 "	25	"	"	"	"
W21	B2 "	4	"	"	"	"
W22	B3 "	21	"	"	"	"

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TABLE XVI. Input capacitance test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
W23	B4	Input capacitance	31	2	15	pF	21
W24	B5	"	13	"	"	"	"
W25	B6	"	30	"	"	"	"
W26	B7	"	9	"	"	"	"
W27	B8	"	37	"	"	"	"
W28	B9	"	38	"	"	"	"
W29	B10	"	19	"	"	"	"
W30	B11	"	16	"	"	"	"

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NOTES:
1. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
2. Null out the effects of the test leads prior to connecting the module.

FIGURE 21. Input capacitance test circuit for table XVI.

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